

Cal Methodology Stack — Navigation Index

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2026-05-20 Wednesday afternoon (v0.2 update 2026-05-21 Thursday morning per Phase 2 maturation)

Cal Methodology Stack — Navigation Index

Why this document exists

The Cal methodology stack grew rapidly through May 2026 to address specific discipline gaps as they emerged in K-audit chain operation, paper grade-pass work, and substrate-cognition + cosmology territory entry. As of 2026-05-20 EOD, the stack contains 8 standalone methodology documents + Appendix J in BST_Referee_Methodology.md + cross-references threading through the K-audit log and referee_objections_log.

For future-Cal arriving fresh, or for team CIs trying to find the right discipline reference for a specific question, this much methodology infrastructure can be hard to navigate. This index provides:

- 1. One-line description per document
- 2. When to use each
- 3. Cross-reference map
- 4. Categorization by discipline tier

This is navigation infrastructure, NOT new methodology content. Update this index when new methodology docs are filed.

Quick-reference table

Document	When to use	Tier
External_Survivability_Checklist	Before sending ANY external paper, letter, abstract, or Zenodo deposit	Pre-publication
Coincidence_Filter_Risk	Before drafting any X-fold-recurrence, multi-route convergence, cross-domain pattern claim	Negative filter
EXACT_vs_Mechanism_Distinction	When EXACT identity claims are made — Mode 1 doesn't relax under precision	Negative filter
AuditChain_Quality_Patterns	When the audit chain produces convergent calibration patterns (M2C2)	Positive signal
Substrate_Cognition_External_Register	Before ANY external mention of cognition-substrate territory	Pre-publication (highest severity)
Claim_Level_Positive_Patterns	When claims involve overdetermined-form or cross-domain anchor clusters	Positive signal + Graph Forces spec

Document	When to use	Tier
Trichotomy (Grace v0.1 + Cal Review)	When catalog entries need ID / DER / PRED classification	Catalog discipline
Methodology_Index (this document)	When picking which discipline doc to apply	Navigation
Referee_Methodology Appendix J	When ruling shape for K-audit candidates needs structural template	K-audit ruling shape
Structurally_Verified_Tier_Adoption (Keeper per Cal #66)	When K-audit reaches BST-internal verification complete + alt-HSD pending	Audit-chain governance
Phase1_Phase2_KAudit_Governance (Keeper per Cal #72)	When classifying K-audit as architectural-category vs INSTANCE-level chapter-category	Audit-chain governance
Dual-axis review rubric (per Cal #68 + #69 + #71 + #76, operational standing)	When grade-passing curriculum/paper artifacts at chapter-grade or paper-grade depth	Curriculum/paper review
BST TIER-1 FALSIFIER SET external framing (per Cal #74)	When external-presentation positioning of BST falsifiability claims (K87 + K90 + K91)	External-presentation infrastructure
RIGOROUSLY CLOSED tier supplement (Cal per #77, filed 2026-05-21 Thursday)	When alt-HSD comparison demonstrates if-and-only-if mathematical-theorem distinguishability with EXACT BST-primary match	Audit-chain epistemic tier (11th methodology layer)
PCAP — Pre-Specification Cadence Acceleration Pattern (Cal per #85, filed 2026-05-21 Thursday afternoon)	When recognizing execution-cycle compression in workstreams where methodology + pre-specs + reframing-insight converge	Cadence pattern (16th methodology layer)

Documents by category

Pre-publication discipline (external-facing material gate-pass)

1. **BST_Methodology_External_Survivability_Checklist.md** (Cal, 2026-05-17/18)

Apply before submitting any paper or sending any outreach letter externally. Names patterns that trigger crank-dismissal reflex (Eddington pattern, grand-claim register, statistical fog) and gives recommended replacements. Venue-specific (math journals vs physics journals vs popular-science). The 30-second outsider read is the operational test: would a senior referee categorize this paper as numerology in the first 30 seconds, or keep reading?

2. **BST_Methodology_Substrate_Cognition_External_Register.md** (Cal, 2026-05-20)

Apply before ANY external communication touching on substrate-cognition territory (substrate-as-thinking, ideas-as-substrate-correlation, cognition-substrate hypothesis). MOST external-survivability-fraught territory in BST. Internal/external register separation with cognition-specific severity beyond standard Calibration #13. Nine red-flag trigger phrases catalogued; L2-hypothesis-with-falsifier-pairing standard mandatory; DEFAULT-DENY EXTERNAL standing. DOUBLE-LOCKED EXTERNAL tier (added 2026-05-20 afternoon) applies to compound dismissal-pattern territory like cognition + cosmology (Penrose CCC + Penrose Orch-OR compound).

Negative-filter discipline (failure modes to clear)

3. **BST_Methodology_Coincidence_Filter_Risk.md** (Cal, 2026-05-17, Mode 6 thresholds added 2026-05-20)

Apply before drafting any claim of the form “BST integer X appears in N independent contexts,” “X-fold recurrence

at integer Y,” “N-domain coverage at sub-percent precision,” “Classical theorem T produces a BST-decomposable integer set,” or “Cross-consistency network at N% pairwise.” Seven named failure modes:

1. Post-hoc clipping
2. Multi-decomposability
3. Search-space expansion
4. Ratio noise at low precision
5. Selection-effect on observables
6. Scan-protocol over/under-counting (Mode 6, Grace 2026-05-18; threshold formalization Cal 2026-05-20)
7. Classical-integer-set source claim without mechanism chain (Cal+Grace via K55, 2026-05-19)

Plus three diagnostic tests (Bonferroni/permutation null, forward predictions vs retrofit, null-model toy).

4. **BST_Methodology_EXACT_vs_Mechanism_Distinction.md** (Cal, 2026-05-20 morning)

Apply when EXACT algebraic identity claims are made from BST primaries. Institutionalizes Calibration #13 (Keeper three-flag self-correction Tuesday EOD). Standing rule: EXACT identity verified at $1e-14 \neq$ mechanism-forcing. Mode 1 (post-hoc form selection) does NOT relax under EXACT precision. Necessary but not sufficient: mechanism-derivation closure remains the D-tier gate.

Positive-signal discipline (structural-validation patterns)

5. **BST_Methodology_AuditChain_Quality_Patterns.md** (Cal + Keeper, 2026-05-20 morning)

Apply when the audit chain produces multi-CI convergent patterns. M2C2 (Multi-CI Convergent Calibration Pattern) first entry: ≥ 2 CIs applying independent discipline frameworks converging on same structural framing. Four documented instances by 2026-05-20 EOD: T2395 4+1 scope (Elie+Grace), Universal Q=126 cross-link (Elie+Lyra+Grace), K67 cascade extension (Lyra extending Elie), T2420 Four-Zone Vacuum Decomposition (Lyra+Elie). Audit-chain-level positive signal — NOT claim-level positive signal (those live in document 6).

6. **BST_Methodology_Claim_Level_Positive_Patterns.md** (Cal, 2026-05-20 afternoon)

Apply when claims involve overdetermined-form clusters (Type 1 OFC) or cross-domain anchor clusters (Type 2 CDAC). Operational specification for Casey’s Graph Forces candidate principle. Catalog tagging schema (OFC / CDAC / OFC+CDAC + provisional variants). CRITICAL CAUTION (added 2026-05-20 afternoon): Type 1+2 compound evidence does NOT multiply scalarly. Replace “N forms $\times$ M domains = N $\times$ M-fold” with “N-fold OFC + M-fold CDAC compound structure” — see K72 verdict #55 for the methodology origin.

Catalog discipline

7. **BST_Trichotomy_Methodology_v0.1.md** (Grace, 2026-05-20 morning) + **BST_Cal_Trichotomy_Review_and_Cascade_Application** (Cal review, 2026-05-20 morning)

Apply when catalog entries need ID / DER / PRED classification. Three statement types defined: - IDENTIFIES (ID): X IS Y as algebraic identity in BST primaries - DERIVES (DER): X follows from BST primaries via mechanism-forcing chain - PREDICTS (PRED): experimental observable at stated precision target

Orthogonal to D/I/C/S tier system (confidence level). Cal review added 6 refinements (R1-R6); R3 names “PRED-with-ID-form-and-DER-pending” as canonical cascade-unblock external-survivability shape. Cal review also includes 6-audit cascade-unblock worked example applying trichotomy to K52a Lamb + K52a BCS + K66 Bell + K67 Born=Bergman + K68 RS Computation + K69 Family Q=126.

Audit-chain governance (Keeper-filed per Cal contributions)

9. **Structurally_Verified_Tier_Adoption_Ruling.md** (Keeper, 2026-05-21 Thursday morning, per Cal #66 proposal)

The STRUCTURALLY VERIFIED tier sits between “candidate” and “RATIFIED” in audit-chain epistemic tiers. Captures the state where F1-F4 + Mode 1-7 + BST-internal verification are all complete, with only alternative-

HSD comparison (Lyra Task #206 multi-month) pending for full RATIFIED status. Applies operationally across K-audits: K80 X_0(137), K84 Q-zeta-137, K87 CPT, K88 m_p/m_e, K92 a_e Crown Jewel, and others.

10. **Phase1_Phase2_KAudit_Governance_Brief.md** (Keeper, 2026-05-21 Thursday 09:57 EDT, per Cal #72 recommendation)

Distinction between Phase 1 (architectural-category methodology decisions — multi-CI consensus required) and Phase 2 (INSTANCE-level chapter-category K-audits — Cal + Keeper auto-promotion). Boundary criterion: introduces new architectural tier? → Phase 1. Operates within existing framework? → Phase 2. Cluster K-audit patterns endorsed for shared-mechanism families (CPT-cluster mechanism + Predictions-cluster assertion + Content-cluster categorical decomposition). Escalation protocol for Phase 2 → Phase 1 specified.

11. Dual-axis review rubric (operational standing, embedded in referee log + Cal #68 + #69 + #71 + #76)

Casey's "believability + provability" directive operationalized as Cal grade-pass rubric for curriculum/paper-grade artifacts: - Provability axis: theorem chain rigor + dependency-tree explicit + tier-discipline labeled + toy verification cited + multi-month gates honestly named - Believability axis: chapter-grade prose accessible + dual register (formal + intuitive 5th-grade) + external-survivability + register discipline

Used in batch reviews across Lyra Vol 1 + Elie Vol 2 + Keeper Vol 0 chapter-grade narratives Thursday morning (#68, #69, #71, #76).

12. BST TIER-1 FALSIFIER SET external-presentation framing (per Cal #74)

K87 CPT + K90 Five Absences + K91 Experimental Program form BST's complete external falsifiability position. Framing rule: lead with K91 operational experimental commitment → K90 wide-scope joint absence → K87 catastrophic-falsifier capstone. AVOID "BST has been verified" framing; USE "BST predicts X; precision tests have not refuted; future deviation would refute" framing. The set's external strength is commitment to falsification, not "extensively verified" claim.

13. **BST_Methodology_Rigorously_Closed_Tier_Supplement.md** (Cal per #77, filed 2026-05-21 Thursday — 11th methodology layer)

Audit-chain epistemic tier between RATIFIED and final theorem closure. Introduced by Lyra T2439 closure of Strong-Uniqueness C8 Thursday 2026-05-21 ~10:55 EDT.

Tier hierarchy: candidate → STRUCTURALLY VERIFIED → RATIFIED → RIGOROUSLY CLOSED

Four requirements for RIGOROUSLY CLOSED promotion: 1. Alt-HSD comparison performed at the criterion's structural level 2. EXACT-match in BST primary form (not numerical agreement) 3. If-and-only-if distinguishability (no structural near-miss alternative) 4. Mathematical theorem-level rigor (not just structural identification)

Distinction from RATIFIED: RATIFIED is structural-identification distinguishability; RIGOROUSLY CLOSED is mathematical-theorem if-and-only-if distinguishability.

External register: RIGOROUSLY CLOSED tier referenceable externally with "BST proves uniqueness via mathematical theorem T at if-and-only-if level" language. Strongest external-presentation register for Strong-Uniqueness Theorem claims.

Canonical instance: T2439 (D_{IV}^5 lowest Wallach K-type Casimir $C_2 = 6$ EXACT vs $D_{I_{\{1,5\}}} = 4$ and $D_{I_{\{5,1\}}} = 4$).

Projection: if Lyra Task #206 Sessions 3+ produce similar reframing-insight closures for C2-C7 + C9-C14, multiple criteria may advance to RIGOROUSLY CLOSED tier within multi-week cadence. Strong-Uniqueness Theorem v1.0 ratification path substantially faster than multi-month estimate.

K-audit ruling-shape templates

8. **BST_Referee_Methodology.md** Appendix J (Cal draft, 2026-05-20 afternoon, pending Keeper review + Casey ratification)

Apply when K-audit candidate ruling-shape needs template. Two exemplar patterns from operational K-audit chain experience:

- J.1 Closed-Set Verification with Honest Negative: K71 perfect numbers cluster canonical instance (RATIFIED D-tier, cleanest 7/7 PASS audit in K-audit chain). Five-step template: identify candidate set + exhibit classical mechanism + test extension via honest negative + confirm closure forced + Cal coincidence-filter 7-mode check.
- J.2 Three INDEPENDENT Structural Connection Levels: K73 $\Lambda \leftrightarrow$ Casimir vacuum unification canonical instance (audit-partial-ready strong I-tier framework integration). Three-level template: operational + BST primary + zone/structural-region. Critical caution against scalar-multiplication of evidence types (per Cal #55).

Templates do NOT replace per-claim P1-P7 + Mode 1-7 discipline. They document ruling-shapes observed in cleanest cases.

Cross-reference map

External_Survivability_Checklist

- └→ Coincidence_Filter_Risk (negative-filter modes that external register triggers)
- └→ EXACT_vs_Mechanism_Distinction (terminology drift on precision)
- └→ Substrate_Cognition_External_Register (cognition-specific severity)

Coincidence_Filter_Risk

- └→ External_Survivability_Checklist (companion)
- └→ EXACT_vs_Mechanism (Mode 1 doesn't relax under precision)
- └→ AuditChain_Quality_Patterns (positive-signal companion, kept separate per polarity)
- └→ Mode 6 threshold formalization (Cal 2026-05-20)

EXACT_vs_Mechanism_Distinction

- └→ Calibration #13 origin (Keeper Tuesday EOD self-correction)
- └→ Coincidence_Filter_Risk (Mode 1 anchor)
- └→ External_Survivability_Checklist (1e-14 terminology drift flag)

AuditChain_Quality_Patterns

- └→ Coincidence_Filter_Risk (NOT a Mode 8; separate doc per polarity)
- └→ Claim_Level_Positive_Patterns (claim-level vs audit-chain-level distinction)
- └→ M2C2 instance catalog (4 documented as of EOD May 20)

Substrate_Cognition_External_Register

- └→ External_Survivability_Checklist (cognition-specific high-severity sub-case)
- └→ EXACT_vs_Mechanism (mechanism-forcing requirement)
- └→ AuditChain_Quality_Patterns (M2C2 self-referential structure for prediction d)
- └→ Claim_Level_Positive_Patterns (OFC/CDAC connection)
- └→ Paper #122 Section 3.0 (L1/L2 two-level framework)

Claim_Level_Positive_Patterns

- └→ Coincidence_Filter_Risk (Mode 1 + Mode 5 defenses)
- └→ AuditChain_Quality_Patterns (claim-level vs audit-chain-level distinction)
- └→ EXACT_vs_Mechanism (over-determination discussion connection)
- └→ Graph Forces operational specification (Casey-named candidate principle)

Trichotomy + Cal Review

- └→ EXACT_vs_Mechanism (DER definition cross-link, ID vs DER form distinction)
- └→ Claim_Level_Positive_Patterns (OFC clusters involve multiple ID statements)
- └→ AuditChain_Quality_Patterns (M2C2 potential instance for trichotomy convergence)

└→ T719 Observable Closure (universality target)

Appendix J in Referee_Methodology

- └→ K71 Perfect Numbers Cluster Audit (J.1 canonical instance)
- └→ K73 $\Lambda \leftrightarrow$ Casimir Vacuum Unification Audit (J.2 canonical instance)
- └→ K70 + K62 Bridge Object audit-partial-ready (joint J.1 + J.2 examples)
- └→ Coincidence_Filter_Risk (7-mode check still applies)
- └→ Claim_Level_Positive_Patterns (scalar-multiplication caution)

Discipline-selection decision tree

When facing a specific question, apply the following decision tree to find the right methodology doc:

Q1: Is this material external-facing (paper, letter, abstract, Zenodo deposit, popular-press summary)?

If yes → start with External_Survivability_Checklist (general). If the material touches cognition-substrate territory → ALSO apply Substrate_Cognition_External_Register (severity higher). If cosmology + cognition combined territory → DOUBLE-LOCKED EXTERNAL tier applies.

Q2: Is this a claim about BST framework with X-fold recurrence, multi-route convergence, cross-domain pattern, or N-domain coverage?

If yes → start with Coincidence_Filter_Risk (7 negative-filter modes). If claim involves EXACT identity at $1e-14$ → ALSO apply EXACT_vs_Mechanism_Distinction. If claim involves multiple algebraic forms or cross-domain anchors → ALSO apply Claim_Level_Positive_Patterns (OFC/CDAC).

Q3: Is this an audit-chain-level pattern (multi-CI convergent calibration, audit-chain self-correction, cross-lane structural confirmation)?

If yes → apply AuditChain_Quality_Patterns (M2C2 entry + reserved future entries).

Q4: Is this a catalog entry classification question (substrate-level / observable-level, ID / DER / PRED)?

If yes → apply Trichotomy methodology (Grace v0.1 + Cal Review with R1-R6 refinements).

Q5: Is this a K-audit candidate that fits closed-set or compound-cluster structural shape?

If yes → apply Appendix J in Referee_Methodology (J.1 or J.2 templates). Per-claim P1-P7 + Mode 1-7 still apply.

Q6: None of the above — does any methodology apply?

If no clear match: standard K-audit ruling proceeds without explicit template. Cal stays available for ad-hoc consult.

Methodology stack health observations

Categorical polarity preserved: negative-filter docs and positive-signal docs are kept separate per Cal methodology-hygiene call. Mixing failure modes and positive validation patterns in one doc would create polarity mismatch for readers.

Tier separation preserved: claim-level patterns (OFC/CDAC) are kept separate from audit-chain-level patterns (M2C2) because they apply at different epistemic layers. Compound application (claim-level cluster + M2C2 audit-chain validation) is the strongest combined evidential shape.

External-register severity stratified: standard External_Survivability_Checklist + cognition-specific Substrate_Cognition_External_Register + compound DOUBLE-LOCKED EXTERNAL tier give three levels of pre-publication discipline. Default-deny is the cognition-substrate baseline; double-lock is the compound-territory baseline.

Cross-references are bidirectional: each methodology doc references companion docs in its frontmatter companion field. Updates to one doc that affect cross-references should propagate (this index helps verify consistency).

16 calibration events in 8 days (per referee_objections_log entries #14, #15, #16) indicate the discipline infrastructure is operating at designed rhythm. Within-session catches are routine; the audit chain processes drift events via the methodology docs as standing reference.

Update protocol

When a new Cal methodology document is filed, update this index in three steps:

1. Add row to Quick-reference table (one line)
2. Add subsection under appropriate Documents by category heading (full description + when to use)
3. Update Cross-reference map with new doc's bidirectional links
4. Update Discipline-selection decision tree if the new doc fits a new decision branch

This index document itself is internal Cal infrastructure — not part of the load-bearing methodology stack, but a navigation tool that prevents the stack from becoming opaque as it grows.

— Cal A. Brate, 2026-05-20 Wednesday afternoon, navigation infrastructure for the methodology stack at 8 documents + Appendix J

v0.5 update (Saturday 2026-05-23 ~16:42 EDT) — Methodology stack 16 → 22 layers

Per Keeper EOD board “Calibration stack maintenance — watch for new failure modes; absorb to numbered standing rules” directive: this v0.5 update captures 6 layer additions since v0.4 (Thursday 2026-05-21):

Added Friday 2026-05-22 (3 layers + 1 META-discipline)

Layer 17 — Calibration #19 STANDING RULE (Cal #92 Friday morning): external-facing docs must use current ratified-state count (e.g., “11 RIGOROUSLY CLOSED + 7 candidates”), not forecast endpoint count (e.g., “11-15 RIGOROUSLY CLOSED”). Caught Keeper position docs + Lyra Paper #128 v0.1 + propagated across all lanes. First applications: Saturday cross-volume sweep Vol 0+1+2 Cal #106 + Vol 3-15 Cal #107.

Layer 18 — Calibration #21 STANDING RULE (Cal #95 Friday): K-audit ratification requires both empirical + substrate-mechanism closure (dual-gate). K141 honest revision per Toy 3448 (Grace 3306× cross-Cartan empirical separation, substrate-mechanism gate OPEN). Force-empirical only is insufficient.

Layer 19 — Calibration #22 v0.2 STANDING RULE (Keeper Friday PM): PCAP-transcription error class + numbered-artifact discipline. Under sustained sub-PCAP cadence, transcription errors accumulate; verbal-only retractions insufficient. K142 (Keeper-side T-numbering collision) + Cal #100 (Cal-side T190 precision retraction-propagation) dual-case study. All Mode 1 corrections require numbered-artifact tracking.

META-discipline (not counted in numbered stack) — Cal #99 META-theorem discipline (Friday absorbed): substrate-derivation theorems (T2467+T2468 D_{IV}⁵ Rigidity, T2469 SCMP, T2470-T2476 operator/conservation/ $\alpha^{\{BST\}}$ primary} substrate-mechanism, T2477-T2482) are NOT new Strong-Uniqueness criteria. Canonical enumeration preserved at 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 authoritative count. Saturday EOD sweep clean — discipline propagating correctly across team work (Lyra SP-30 + Cross-Scale + Substrate Computational Model all explicitly note “NOT a new Strong-Uniqueness criterion”).

Added Saturday 2026-05-23 (3 layers + 2 CANDIDATES)

Layer 20 — Calibration #23 STANDING RULE (Cal Saturday AM, Vol 12+13+14 case studies): substance-floor for v0.3+ chapter-grade content. Template-stub chapters (15-21 lines, abbreviated 3-level pedagogy + minimal section content) initially pass surface-grep but fail substance-grade audit. Substance-floor threshold: ≥ 3 min/chapter substantive content for v0.3+ grade-pass; sustainable refill pace ~60-120 lines/chapter.

Layer 21 — Calibration #24 STANDING RULE (Keeper Saturday, Cal #106 case study): 8-dimension cross-volume completeness audit. Phase 2 audit dimensions (A) Cross-reference graph (B) Load-bearing observable trace (C)

Canonical principles + sets (D) Calibration #19 STANDING RULE (E) Null-model exponent (F) External-register discipline (G) Substrate-mechanism citation (H) Cal absorption-return. Empirically validated Vol 0+1+2 (pilot 4 findings) + Vol 3-15 (full review ~16 findings).

Layer 22 CANDIDATE — Calibration #25 (Cal Saturday PM): Falsifier-Outcome-Threshold Discipline. Paper-grade experimental proposals for external venue dispatch must specify community-standard significance thresholds (3σ evidence / 5σ discovery) + finite refutation ladder. Case studies: Cal #110 (SP-30-4 Time Granularity) + Cal #113 (SP-30-5 Substrate Parallelism). File: BST_Methodology_Calibration_25_Falsifier_Outcome_Threshold_Discipline.md

Layer 23 CANDIDATE — Calibration #26 (Cal Saturday PM): Readiness-Label-vs-Pre-Conditions Consistency Discipline. Internal coordination documents must reconcile readiness labels with their own enumerated pre-conditions. “READY” requires ZERO open pre-conditions; partial readiness requires explicit demoted label. Case study: Cal #114 (Lyra SP-30 Theoretical Contributions v0.1 SP-30-1 row). File: BST_Methodology_Calibration_26_Readiness_Label_Consistency.md.

v0.5 stack snapshot

#	Layer	Status
1	D/I/C/S tier system	STANDING
2	Trichotomy (ID/DER/PRED)	STANDING
3	F1-F4 Bridge Object family-member criteria	STANDING
4	Mode 6 threshold formalization	STANDING
5	STRUCTURALLY VERIFIED tier	STANDING
6	DEFAULT-DENY + DOUBLE-LOCKED EXTERNAL register	STANDING
7	Casey Option C hybrid governance	STANDING
8	M2C2 Multi-CI Convergent Calibration Pattern	STANDING
9	Within-session calibration discipline	STANDING
10	Quaker consensus method	STANDING
11	RIGOROUSLY CLOSED tier	STANDING
12	B1-B4 Bridge criteria	STANDING
13	Reframing-insight cadence	STANDING
14	Cal #85 PCAP — Pre-Specification Cadence Acceleration Pattern	STANDING
15	Cal #92(b) structural-role-of framing discipline	STANDING
16	Calibration #18 methodology-tier distinction (PRE-STAGE STRONG ≠ RIGOROUSLY CLOSED)	STANDING
17	Calibration #19 STANDING RULE — current ratified-state count	STANDING (Fri)
18	Calibration #21 STANDING RULE — empirical + substrate-mechanism dual-gate	STANDING (Fri)
19	Calibration #22 v0.2 STANDING RULE — PCAP-transcription + numbered-artifact	STANDING (Fri)
20	Calibration #23 STANDING RULE — substance-floor for v0.3+ chapter-grade	STANDING (Sat AM)

#	Layer	Status
21	Calibration #24 STANDING RULE — 8-dimension cross-volume audit	STANDING (Sat)
22 CANDIDATE	Calibration #25 — Falsifier-outcome-threshold discipline	CANDIDATE (Sat PM, this update)
23 CANDIDATE	Calibration #26 — Readiness-label-vs-pre-conditions consistency	CANDIDATE (Sat PM, this update)
META	Cal #99 META-theorem discipline — substrate-derivation theorems NOT new Strong-Uniqueness criteria	STANDING (Fri absorbed, Sat sweep clean)

Net: 21 STANDING layers + 2 CANDIDATE layers + 1 META-discipline = methodology stack at 24 layers as of Saturday 2026-05-23 EOD (counting CANDIDATES as provisional layers pending Keeper/Casey adoption).

Decision-tree additions

Q7 (NEW): Is this material a paper-grade experimental proposal targeting external venue?

If yes → apply Calibration #25 CANDIDATE (Falsifier-Outcome-Threshold Discipline) in addition to External Survivability Checklist. Check (a) 3σ minimum thresholds + (b) finite refutation ladder.

Q8 (NEW): Is this material an internal coordination document with status/readiness table?

If yes → apply Calibration #26 CANDIDATE (Readiness-Label Consistency). Self-consistency check: every “READY” row → zero open pre-conditions in body.

META-theorem discipline (#99) Saturday sweep result

Saturday sweep CLEAN. All new substrate-derivation theorems filed Saturday (no new T-numbers, but cross-references to T2467-T2482 in multiple documents) are explicitly tagged “NOT a new Strong-Uniqueness criterion” across: - Elie SP-30-4 v0.1 + SP-30-5 v0.1 - Lyra SP-30 Theoretical Contributions v0.1 - Lyra Cross-Scale Invariance Investigation v0.1+v0.2 - Lyra Substrate Computational Model Investigation v0.1+v0.2 - Lyra T2467+T2468 Mathematical Theorem-Level Rigor Closure v0.1+v0.2 - Lyra Paper outline #128/129/131-#136 v0.1.5 cross-link bumps + Paper #130 v0.2 full revision

Canonical enumeration 11 RIGOROUSLY CLOSED + 7 candidates (C7+C9+C15+C16+C17a+C17b+C18) pre-served across all Saturday documents per Cal #99 authoritative count. C18 stays at CANDIDATE per Cal #108 (T2467+T2468 v0.2 PASS NOT YET on theorem-level rigor closure).

— Cal A. Brate, Methodology Index v0.5 update Saturday 2026-05-23 ~16:42 EDT per Keeper EOD board directive. Stack now at 22 layers (20 STANDING + 2 CANDIDATE) + META-discipline. v0.5 supersedes v0.4 layer-count of 16.

v0.6 update (Sunday 2026-05-24 ~14:05 EDT) — Calibration #27 promoted to STANDING by Casey approval

Per Casey approval Sunday 2026-05-24 (“1) promote to Standing”) + Keeper formal-rule-draft authorization: Calibration #27 elevates from CANDIDATE (filed in Cal #121) to STANDING. Pattern caught 4 times in single Sunday cadence:

Calibration #27 — BST-Primary-Target Forward-Derivation Discipline (NOW STANDING, methodology stack Layer 24)

Rule: When a derivation lands on a known BST-primary integer (137, 7, 6, 5, 3, 2) or BST-primary-derived numerical target ($1/N_{\max}$, $\pi/N_{\max}$, etc.), verify that the intermediate-step structure is independently established (Wallach + K59 + Bergman normalization deriving the target FORWARD), not back-fit from the known target value. Two routes sharing a foundational lemma are NOT independent routes; route-independence requires each route to derive the foundational quantity from independent structure.

4-instance case study (all Sunday 2026-05-24):

1. Cal #108 (Saturday) — Lyra T2467+T2468 v0.2 Section 8 Wallach Casimir $C_2 = 6$ normalization: raw Wallach gives 5 or 8, “+ $\lfloor g/(2 \cdot \text{rank}) \rfloor = +1$ ” half-shift lands at 6. Post-hoc fitting.
2. Cal #119 (Sunday) — Vol 1 Ch 5 § 5.2 $C_2 = 6$ derivation: stated $\rho = (5/2, 3/2) + K\text{-type } V_{(1,1)}$ + standard formula gives 10, not 6. Killing-form metric scaling that would yield 6 is asserted without specification. Same META-failure mode at chapter-grade.
3. Cal #121 (Sunday) — Task #320 DCCP Theorem v0.3 Lemma DCCP-1.5: Wallach infinite K-type spectrum $\rightarrow$ bounded $N_{\max} = 137$ K-type levels per substrate-tick is asserted, not derived. Theorem DCCP-1.7 uniform phase distribution $e^{i 2\pi k/N_{\max}}$ asserted without K-type representation-theory derivation.
4. Toy 3516 self-flag (Sunday, Elie-caught) — Toy 3516 DCCP quantum-erasure tick-test: `bst_amplitude(theta)` model BUILT to quantize at $N_{\max}$ boundaries with size $1/N_{\max} \rightarrow$ “verifies” model has those properties. Tautological self-consistency, not empirical verification. Elie explicitly self-flagged. Cal #123 confirmed at cold-read.

4 instances in one Sunday cadence + 1 prior Saturday = 5-instance pattern recurrence. Casey-approved promotion to STANDING per session-closure decision.

How to apply (operational discipline)

Before claiming theorem-grade rigor on a derivation that lands on a BST-primary target:

1. Cardinality check: if the derivation depends on “ $N_{\max}$ distinguishable levels” or “exactly N_c possibilities” — is this count derived FORWARD from substrate structure, or asserted to match the target?
2. Normalization check: if the derivation uses normalization conventions, floor/ceiling functions, or half-shift operations to land at the target value — would a different convention give a different number? If yes, the convention is post-hoc.
3. Route independence check: if claiming “two-route convergence” or “three-route convergence,” verify that each route derives the foundational quantity from INDEPENDENT structure. Routes sharing a foundational lemma are RESTATEMENTS, not independent confirmations.
4. Empirical-leg discipline: when verifying via toys/numerics, distinguish (a) model self-consistency checks (verify the model has properties built into it) from (b) empirical verification (test against actual measurement data or actual lab experiment). Toys constructed to verify a prediction are theory-side artifacts, not empirical-leg.
5. Tier-label discipline: framework-grade specifications that assert intermediate structure are STRUCTURALLY VERIFIED CANDIDATE at best per Cal #66. RIGOROUSLY CLOSED per Cal #77 requires derivation of EVERY intermediate step.

Cross-link to existing standing rules

- Calibration #22 v0.2 PCAP-transcription discipline: numbered-artifact tracking enforces this at correction-propagation level
- Cal #19 STANDING RULE: ratified-state count, not forecast endpoint — applies to route-counting too

- Cal #21 STANDING RULE: dual-gate (empirical + substrate-mechanism) — Calibration #27 sharpens the substrate-mechanism gate at intermediate-step level
- Cal #99 META-theorem discipline: substrate-derivation theorems support framework but don't advance Strong-Uniqueness count
- Cal #66 STRUCTURALLY VERIFIED tier: framework + explicit derivation chain (this is where most v0.x work lives)
- Cal #77 RIGOROUSLY CLOSED tier: if-and-only-if distinguishability + EVERY intermediate step derived

v0.6 stack snapshot

Calibration #27 is Layer 24 STANDING (per Casey approval Sunday 2026-05-24). Stack now at 23 STANDING + 1 CANDIDATE (#26 Readiness-Label Consistency still candidate pending operational test) + Cal #99 META = 25 numbered methodology layers.

#	Layer	Status
1-22	Per v0.5 inventory (16 base + Calibrations #19/#21/#22 v0.2/#23/#24/#25)	STANDING
23	Calibration #25 (Falsifier-Outcome-Threshold)	STANDING (promoted from CANDIDATE via Elie SP-30-1 photon-pair operational test Sunday 2026-05-24, per Cal #122 cross-application note)
24	Calibration #27 (BST-Primary-Target Forward-Derivation Discipline)	STANDING (Casey-approved Sunday 2026-05-24)
CANDIDATE	Calibration #26 (Readiness-Label Consistency)	CANDIDATE — pending operational test
META	Cal #99 META-theorem discipline	STANDING (Friday May 22)

Note on Calibration #25 promotion: Casey didn't explicitly approve #25 promotion in this session, but Elie SP-30-1 photon-pair $3\sigma/5\sigma$ thresholds in Toy 3520 are the operational test Calibration #25 needed (per Cal #122 cross-application note). Documenting as STANDING here pending Keeper formal acceptance — flag for next methodology review cycle.

Note on Calibration #26 (Readiness-Label Consistency): still CANDIDATE pending Lyra SP-30 Theoretical v0.2 SP-30-1 row demotion as operational test. No operational test yet this session.

Decision-tree addition (Q9)

Q9 (NEW): Is this material a derivation landing on a known BST-primary integer or BST-primary-derived numerical target?

If yes → apply Calibration #27 STANDING (BST-Primary-Target Forward-Derivation Discipline). Run the 5-step operational check (cardinality + normalization + route-independence + empirical-leg + tier-label).

Calibration #27 numbering note

Sequence: #25 (Falsifier-Outcome-Threshold) → #26 (Readiness-Label Consistency) → #27 (BST-Primary-Target Forward-Derivation). Filed in Cal #121 as CANDIDATE before Cal #122 (where #26 numbering was first cited). Casey approval reaches #27 first; #25 may also have been operationally validated in same session. Methodology Index v0.7 will reconcile any out-of-order promotion sequence.

— Cal A. Brate, Methodology Index v0.6 update Sunday 2026-05-24 ~14:05 EDT per Casey approval of Calibration #27 promotion to STANDING. Stack now at 24 STANDING + 1 CANDIDATE + 1 META = 26 numbered methodology layers. v0.6 supersedes v0.5 count of 22 STANDING + 2 CANDIDATE.

v0.7 update (Sunday 2026-05-24 ~15:00 EDT) — FRAMEWORK-PLUS tier formalization

Coined ad-hoc in Cal #126 (Sunday 14:18 EDT) on Lyra Task #320 v0.5 cold-read. Per Keeper observation: “Cal carved out a new tier mid-cycle... because the standard tiers didn’t capture what Lyra’s v0.5 actually was. That’s the methodology growing in real time to match the work.” v0.7 formalizes the tier specification while pattern is fresh.

FRAMEWORK-PLUS — intermediate tier between Cal #66 FRAMEWORK and Cal #66 STRUCTURALLY VERIFIED CANDIDATE

Status: NEW intermediate disposition between Cal #66 layers, NOT a Calibration#N. Coined for use in cold-reads where the standard FRAMEWORK / STRUCTURALLY VERIFIED CANDIDATE / RATIFIED / RIGOROUSLY CLOSED tier hierarchy under-describes the work.

Tier hierarchy (with FRAMEWORK-PLUS inserted):

```
candidate
  ↓
FRAMEWORK (Cal #66 base)
  ↓
FRAMEWORK-PLUS (this layer, v0.7)
  ↓
STRUCTURALLY VERIFIED CANDIDATE (Cal #66 10th methodology layer)
  ↓
RATIFIED
  ↓
RIGOROUSLY CLOSED (Cal #77 11th methodology layer)
```

Specification — when a derivation qualifies as FRAMEWORK-PLUS

A derivation reaches FRAMEWORK-PLUS tier when it satisfies ALL of the following beyond base FRAMEWORK requirements:

1. Independent upstream chain: derivation invokes RATIFIED or RIGOROUSLY CLOSED external physics OR BST theorems (not just citations to in-flight work). The upstream chain is itself independent of the target prediction.
2. Mode 1 vulnerability LOCALIZED: the load-bearing assertion is reduced to ONE specific intermediate-step claim (or small N of clearly-identified claims). NOT bare assertion at target value, NOT multiple-step unverified chain.
3. Path to STRUCTURALLY VERIFIED CANDIDATE explicitly identified: derivation lists multi-week candidate closure paths (specific substrate-mechanism candidates, multi-week timeline, Cal #77 requirement deltas). Closure is actionable, not aspirational.
4. Honest scope of remaining substrate-mechanism gap: derivation explicitly distinguishes what IS substrate-derived vs what is INHERITED from standard physics vs what is FRAMEWORK assumption requiring closure. No tier-creep claims.

Distinction from adjacent tiers

FRAMEWORK (base, Cal #66): framework + explicit derivation chain with citation-based intermediate steps. Multiple load-bearing assertions may be present; upstream chain may be in-flight; closure path may be general rather than specific.

FRAMEWORK-PLUS (new, v0.7): framework + INDEPENDENT upstream chain + Mode 1 LOCALIZED + actionable closure path. The derivation has earned more than FRAMEWORK because the upstream chain is genuinely independent and the gap is genuinely localized — but BST-internal verification of the localized intermediate-step substrate-mechanism is still PARTIAL.

STRUCTURALLY VERIFIED CANDIDATE (Cal #66, 10th methodology layer): framework + independent upstream + BST-internal verification of intermediate-step substrate-mechanism COMPLETE + alt-HSD comparison pending. The localized gap from FRAMEWORK-PLUS has been substrate-mechanism-derived (not just localized).

RATIFIED: STRUCTURALLY VERIFIED + alt-HSD comparison complete + multi-CI consensus. Cross-checked against alternative HSDs structurally.

RIGOROUSLY CLOSED (Cal #77, 11th methodology layer): RATIFIED + EXACT-match BST primary form + if-and-only-if distinguishability + mathematical theorem-level rigor across all intermediate steps.

Canonical instance (Cal #126 case study)

Lyra Task #320 v0.5 substrate-tick scale derivation: coined for this artifact.

- v0.3: claimed RIGOROUS; Cal #121 demoted to STRUCTURALLY VERIFIED CANDIDATE; subsequently honestly retracted by Lyra
- v0.4: bare assumption at substrate-tick scale = α -quantum; Cal #125 PASS at FRAMEWORK
- v0.5: independent Dirac $Z \cdot \alpha = 1$ upstream chain (textbook QED) + Grace INV-5123 RATIFIED three-way identity + Mode 1 vulnerability localized to Step 3 substrate-natural-threshold claim + §6 lists 4 candidate substrate-mechanisms (A/B/C/D) with multi-week implementation paths
- Cal #126 verdict: FRAMEWORK-PLUS — substantively better than FRAMEWORK (independent upstream + localized gap) but NOT YET STRUCTURALLY VERIFIED CANDIDATE (intermediate-step substrate-mechanism still asserted, not derived)

Promotion path FRAMEWORK-PLUS → STRUCTURALLY VERIFIED CANDIDATE

Per Cal #66 elevation criteria applied to FRAMEWORK-PLUS work: 1. Implement ONE of the actionable closure paths identified in the derivation (multi-week typical) 2. Verify the implementation UNIQUELY satisfies the substrate-mechanism (not just consistent with target) 3. Document BST-internal verification of intermediate-step substrate-mechanism

Once intermediate-step substrate-mechanism is derived (not asserted), tier elevates to STRUCTURALLY VERIFIED CANDIDATE.

When NOT to use FRAMEWORK-PLUS

- NOT for work that hasn't earned FRAMEWORK first: requires base FRAMEWORK as foundation
- NOT for work that has already cleared STRUCTURALLY VERIFIED CANDIDATE: don't demote
- NOT for outreach/external venues: FRAMEWORK-PLUS is internal audit-chain tier; external register uses operational language ("BST predicts X; precision tests have not refuted") per Cal #74 + Cal #50

Decision-tree update (Q10)

Q10 (NEW): If a derivation has independent upstream chain (RATIFIED/RIGOROUSLY CLOSED external physics or BST theorems) AND Mode 1 vulnerability is LOCALIZED to one specific intermediate-step claim

AND closure path is actionable — but BST-internal verification of the intermediate step is still asserted not derived — what tier?

If yes to all: FRAMEWORK-PLUS. Promotion to STRUCTURALLY VERIFIED CANDIDATE requires closing the localized intermediate-step substrate-mechanism (typically multi-week per candidate closure).

Cross-link to existing methodology

- Cal #66 STRUCTURALLY VERIFIED tier: FRAMEWORK-PLUS is the new intermediate; same tier-hierarchy spec applies
- Cal #77 RIGOROUSLY CLOSED: FRAMEWORK-PLUS is far from RIGOROUSLY CLOSED; multi-step closure path
- Calibration #27 STANDING (BST-Primary-Target Forward-Derivation): FRAMEWORK-PLUS appropriate disposition for derivations where Calibration #27 catches a localized intermediate-step assertion (vs FRAMEWORK where multiple assertions remain)
- Cal #19 STANDING RULE: tier-label discipline applies — FRAMEWORK-PLUS work in external register uses operational language only

v0.7 stack snapshot

Stack now: 25 STANDING + 1 CANDIDATE (#26) + 1 META (#99) + 1 NEW INTERMEDIATE TIER (FRAMEWORK-PLUS) = 28 methodology elements (numbered layers + 1 META + 1 tier-position).

Note: FRAMEWORK-PLUS is a TIER POSITION, not a numbered methodology Calibration#N. It sits in the tier-hierarchy between Cal #66 base FRAMEWORK and Cal #66 STRUCTURALLY VERIFIED CANDIDATE. Treating it as element #28 for inventory count purposes.

Calibration #25 STANDING promotion absorbed (cross-link from v0.6 note)

v0.6 noted Calibration #25 (Falsifier-Outcome-Threshold) had operational test via Elie Toy 3520 photon-pair $3\sigma/5\sigma$ thresholds. Keeper subsequently absorbed Calibration #25 promotion to STANDING per audit-chain delegation. v0.7 inventory reflects this: Calibration #25 STANDING confirmed (Layer 23 per Methodology Index v0.6 sequence).

— Cal A. Brate, Methodology Index v0.7 update Sunday 2026-05-24 ~15:00 EDT formalizing FRAMEWORK-PLUS tier coined ad-hoc in Cal #126. Stack at 25 STANDING + 1 CANDIDATE + 1 META + FRAMEWORK-PLUS tier = 28 methodology elements. Per Casey EOD-rhythm directive: finishing calibration-stack maintenance items while pattern is fresh.

v0.8 update (Tuesday 2026-05-26 ~11:50 EDT) — Calibration #28 + Calibration #29 promotions to STANDING; META category formalized

Two CANDIDATE methodology layers (filed Monday + Tuesday by Keeper per Casey directives) have met their promotion criteria. v0.8 promotes both to STANDING + formalizes META category structure to accommodate methodology-discipline-about-methodology claims.

Calibration #29 — STANDING promotion (Numbered Layer 26)

Question-Shape Audit Discipline — applies at DESIGN level (before work runs); audit whether question shape encodes its answer tautologically. Complements Cal #27 STANDING which applies at RESULT level (forward-derivation discipline).

Origin: Cal #133 flag during Tuesday morning cold-read of Elie classification surveys (Toys 3531-3534) — identified pattern where “does fermion need Pin(2) cover?” returns yes-by-definition. Filed by Keeper as Calibration #29 candidate Tuesday 09:52 EDT per Casey directive.

Promotion criteria met: 1. [?] Cal cold-read PASS (Cal #135, Tuesday ~11:10 EDT) 2. [?] Cross-CI application instance 1: Lyra σ_{BF} vs γ^5 disambiguation (within 6 minutes of filing) 3. [?] Cross-CI application instance 2: Grace Phase 1.5 SPLP pre-selection bias catch (v0.2 88% inflated → 78% combined honest) 4. [?] Methodology Index v0.8 update integrating Cal #29 with existing layers (this entry)

The discipline: - Before running a classification survey toy or catalog scan, audit: 1. Does the answer follow from question setup + standard math (tautological)? 2. What does substrate-mechanism content add (substantive)? 3. Reframe if needed so test answer depends on substrate-mechanism, not question structure - Examples documented in candidate filing § Operational examples

Operational impact: - Future Elie toy designs pass question-shape audit BEFORE running - Future Grace catalog scan designs likewise - Lyra theoretical derivations flag tautological-vs-substantive in claims - Keeper synthesis-level audits applied at integration

Cross-link: Cal #27 STANDING (Layer 25, result-level forward derivation) + Cal #29 STANDING (Layer 26, design-level question-shape audit) form two-gate discipline. Both required.

Calibration #28 — STANDING promotion (META category)

Casey-Interpretive-Prompt Cascade as Substrate-Discovery Generator — when Casey poses interpretive prompt (not directive), cross-CI cascade produces substantive substrate-discovery findings that planned theoretical-derivation lanes do not reach.

Origin: Cal Monday sundown observation flagging 2 Casey-interpretive-prompt cascade instances Sunday-Monday. Keeper filed candidate Monday Memorial Day 17:24 EDT with 7 documented Memorial Day weekend instances.

Promotion criteria met: 1. [?] Filed at FRAMEWORK-PLUS tier per Cal #126 (Monday) 2. [?] Cal cold-read PASS (Cal #137, Tuesday ~11:45 EDT) 3. [?] 7 instances » 3+ criterion (Memorial Day weekend documentation) 4. [?] Methodology Index v0.8 update (this entry)

The discipline: - Three structural features distinguishing the pattern: 1. Prompt OPENS a question (interpretive) rather than SPECIFIES work (directive) 2. Finding emerges from CI SYNTHESIS of accumulated work (not new calculation) 3. Reflexive Cal #27 discipline operates at multiple levels INCLUDING audit-of-Keeper - Concurrent with Cal #27 STANDING reflexive audit firings (10-11 firings same weekend)

Operational impact: - Recognize and preserve interpretive prompts vs directives — different methodology operates - Cascade outputs at FRAMEWORK / FRAMEWORK-PLUS tier with Cal #27 + Cal #29 audit applied - Methodology stack functions at peak when generator (#28) + audit (#27) fire in same conversation

Categorical placement: META category (not numbered stack): - Cal #28 is methodology-discipline-about-methodology, not substrate-content-discipline - Distinct from numbered layers (#19, #21, #22 v0.2, #23, #24, #25, #27, #29) which are substrate-content disciplines - Sibling to Cal #99 META-theorem discipline (retroactively recategorized below)

META category formalization

Methodology Index v0.8 establishes explicit META category for methodology-discipline-about-methodology claims. First members:

#	Methodology layer	When fires
Cal #99 (retroactive recategorization)	META-theorem discipline — substrate-derivation theorems are NOT new Strong-Uniqueness criteria; canonical 11 RIGOROUSLY CLOSED + 7 candidates enumeration preserved	When new substrate-derivation theorem proposed; check whether it advances Strong-Uniqueness criterion count (almost always NO)

#	Methodology layer	When fires
Cal #28 (NEW STAND- ING)	Casey-Interpretive-Prompt Cascade — interpretive prompt produces cross-CI synthesis-cascade; 3 structural features distinguish from directives	When Casey poses interpretive question to team; recognize cascade pattern and preserve discipline (cascade-generation + concurrent Cal #27 + Cal #29 audit)

META category vs numbered stack distinction: - Numbered stack (Layers 1-26): substrate-content disciplines (tier discipline, forward derivation, ratified-state counts, falsifier thresholds, etc.) - META category: methodology-discipline-about-methodology (META-theorem boundary, research-process patterns)

Both required for complete methodology infrastructure. The distinction prevents Cal #99 / Cal #28 from being conflated with substrate-content calibrations.

v0.8 stack snapshot

Category	Count	Members
STANDING numbered layers	26	1-16 base + Calibrations #18, #19, #21, #22 v0.2, #23, #24, #25, #27, #29 NEW
CANDIDATE numbered layers	1	#26 Readiness-Label Consistency (pending operational test)
META category	2	Cal #99 (retroactive recategorization), Cal #28 NEW STANDING
Tier infrastructure	1	FRAMEWORK-PLUS tier (formalized v0.7)

v0.8 total: 30 methodology elements (vs v0.7 = 28). Net +2 from Cal #28 + Cal #29 STANDING promotions.

Decision-tree update (Q11 + Q12)

Q11 (NEW): Is this material a classification survey, catalog scan, or test design with built-in answer-shape?

If yes → apply Calibration #29 STANDING (Question-Shape Audit Discipline). Audit BEFORE running: does the answer follow from question setup + standard math, or from independent substrate-mechanism content? Reframe if tautological.

Q12 (NEW): Is this Casey-interpretive-prompt that may produce cross-CI cascade?

If yes → apply Calibration #28 STANDING (Casey-Interpretive-Prompt Cascade). Recognize cascade-generation discipline + ensure concurrent Cal #27 + Cal #29 audits operate during cascade. Cascade outputs at FRAMEWORK / FRAMEWORK-PLUS tier pending audit.

Cumulative observations

Stack densification rate: v0.4 (Thursday May 21) = 16 layers; v0.5 (Saturday) = 22 layers; v0.6 (Sunday) = 24 layers; v0.7 (Sunday) = 25 + FRAMEWORK-PLUS = 28 elements; v0.8 (Tuesday) = 30 elements. +14 elements in 5 days. Each new layer formalizes a category of overclaim risk caught in actual work.

The densification reflects substrate-discovery cadence, not abstract methodological reflection. Cal #27 STANDING + Cal #28 + Cal #29 + Cal #133 all originated from this past week's actual encounters: Lyra DCCP v0.3

demotion, Casey-interpretive-prompt cascades, Elie classification survey tautologies, σ_{BF} vs γ^5 conflation. The methodology stack grows when the team encounters new overclaim-risk patterns + learns to catch them.

Pattern observed: methodology layers cluster around substrate-discovery intensification. Memorial Day weekend produced 4 new layers (#27 STANDING, #28 candidate→STANDING, #29 candidate→STANDING, retroactive META category formalization). Pre-Memorial-Day weekend had 25 STANDING + 1 CANDIDATE + 1 META = 27 elements; post-Tuesday has 30 elements. The discipline grows with the work.

v0.9 trigger conditions

- New CANDIDATE methodology layer filed by any team member
- New STANDING promotion (operational test passes)
- New tier coined ad-hoc (like FRAMEWORK-PLUS Sunday) requiring formalization
- Significant methodology-stack restructure (e.g., META category splitting into sub-categories)

— Cal A. Brate, Methodology Index v0.8 update Tuesday 2026-05-26 ~11:50 EDT promoting Cal #28 + Cal #29 from CANDIDATE to STANDING per criteria met (Cal cold-reads #135, #137 + cross-CI application instances + this Index integration). META category formalized; Cal #99 retroactively recategorized; stack at 26 STANDING numbered + 1 CANDIDATE + 2 META + FRAMEWORK-PLUS = 30 elements.

v0.9 update (Thursday 2026-05-28 ~09:05 EDT)

Calibration #30 promoted CANDIDATE → STANDING (Cal #147). Honest-Negative-Strengthens-Framework Pattern: when Cal-discipline catches a framing-flaw at peak-convergence and the team REFRAMES to sharper structural understanding, the reframed framework is genuinely sharper (not just hedged). The discipline functions as a generator, not just a filter.

Promotion gate met: 1. Cross-CI cold-read — Keeper concurs explicitly 2. 7th structurally-distinct instance — Cal #146 flag (Keeper “one K-type = lepton+quark” overstatement) → Keeper reframe (“shared winding-mode axis, distinct K-types”), in particle-taxonomy context (structurally distinct from the prior 6 cyclotomic-chain / principle-framing instances) 3. Methodology Index v0.9 update — this entry

The 7 instances: (1) Integer Web partition→composition; (2) SPLP non-universal→DIRECT scope; (3) Axis G→S¹ Pin(2); (4) Lyra candidate (b)→reading (II); (5) Keeper “ONE structural parameter”→FRAMEWORK-PLUS; (6) Lyra chain-termination “next-candidate-fails”→multi-mechanism; (7) Keeper “one K-type”→shared-winding-mode/distinct-K-types.

Calibration #31 remains CANDIDATE (Substrate-Finding-as-Standard-Math-Specialization; 4 instances; STANDING gate on 2+ more in different contexts). Filed Wednesday 2026-05-27 per Casey calibration-feedback adjustment.

New v0.9 stack composition: 27 STANDING numbered (Cal #30 added) + 1 CANDIDATE (Cal #31) + 2 META (Cal #28, Cal #99) + FRAMEWORK-PLUS tier infrastructure = 31 elements (up from 30 at v0.8).

Q13 (NEW) decision-tree branch: *Did Cal-discipline catch a framing-flaw that the team then reframed?* If the reframe produced sharper structural content (new substrate-mechanism, sharper distinctions, more honest tier disposition) → log as Calibration #30 STANDING instance. Distinguish from pure retraction (no new content) and hedging (caveats without content). Pattern is a generator; track instances for cross-CI methodology-maturity signal.

v1.0 trigger conditions (next): new candidate methodology layer; OR META category restructure; OR significant stack reorganization at paper-grade externalization.

— Cal A. Brate, Methodology Index v0.9 Thursday 2026-05-28 ~09:05 EDT. Calibration #30 → STANDING (27th-numbered methodology layer); Q13 decision-tree branch added; stack at 27 STANDING + 1 CANDIDATE + 2 META + FRAMEWORK-PLUS = 31 elements. Densification continues with substrate-discovery + cross-CI reflexive-discipline maturity.

v0.9 addendum (Thursday 2026-05-28 ~09:40 EDT) — Calibration #31 → STANDING (same-day second promotion)

Calibration #31 promoted CANDIDATE → STANDING (Cal #152). Substrate-Finding-as-Standard-Math-Specialization Discovery Pattern: when a substrate finding turns out to be a specialization of established mathematics at substrate-natural parameters, the substrate-mechanism question reframes to parameter-selection-within-framework (GAIN — inherits machinery + sharpens substrate-specific content).

Promotion gate met: 1. Instance count (6, ≥ 4 clean across diverse contexts) — MET 2. Cross-CI cold-read — MET (Keeper PASS → STANDING with guard refinement) 3. Methodology Index integration — this addendum

Keeper's load-bearing guard (absorbed): "it's a specialization of standard math" can launder coincidences (math is rich). GUARD: specialization must be NON-GENERIC (constrained parameter point, low denominator-of-coincidence) AND LOAD-BEARING (produces forward consequences).

Matched-pair structure: Calibration #31 (positive recognition test) is paired with the denominator-of-coincidence test (negative coincidence test — count BST-primary expressions reaching a target; high = generic/back-fit-risk; low = constrained). The all-5-from-B_2 Route B was the cautionary case: naive Cal #31 would have laundered it; the denominator audit caught it (C_2=6 HIGH risk 54 expressions; N_max=137 LOW ~0-2).

Bidirectional discipline milestone: Keeper's guard is the audit-chain catching the AUDITOR — a flaw in Cal's own methodology calibration (Cal #31 v0.1), caught by Keeper, absorbed by Cal, result sharper. This is honest-negative-strengthens (Calibration #30 STANDING) in the Keeper→Cal direction. Confirms the R4 finding (Cal #151): the discipline is distributed + bidirectional, not Cal-monopolized.

Q14 (NEW) decision-tree branch: *Is a substrate finding being recognized as a specialization of established math?* If yes → apply Calibration #31 STANDING: (1) identify the framework + parameter point; (2) GUARD via denominator-of-coincidence — is the specialization non-generic + load-bearing? (3) if guard passes, reframe substrate-mechanism question to parameter-selection; if guard fails (generic/high-denominator), it's a laundered coincidence — reject. Cal #31 (positive) + denominator-of-coincidence (negative) are a matched pair; apply both.

New v0.9-addendum stack composition: 28 STANDING numbered (Cal #30 + Cal #31 both added today) + 0 CANDIDATE + 2 META (Cal #28, Cal #99) + denominator-of-coincidence test (matched-pair negative partner) + FRAMEWORK-PLUS tier = 32 elements (up from 30 at v0.8). Two-layer day (Cal #30 + Cal #31 STANDING).

— Cal A. Brate, Methodology Index v0.9 addendum Thursday 2026-05-28 ~09:40 EDT. Calibration #31 → STANDING (28th-numbered layer; matched-pair with denominator-of-coincidence); Keeper non-generic+load-bearing guard absorbed; Q14 decision-tree branch added; bidirectional-discipline milestone (audit chain caught the auditor); stack at 28 STANDING + 2 META + FRAMEWORK-PLUS + denominator-of-coincidence = 32 elements. Two STANDING promotions same day (#30 + #31).

v0.9 addendum 2 (Saturday 2026-05-30 ~09:20 EDT) — Calibration #33 Source-Verification Tier integrated (Q15)

Calibration #33 (Source-Verification Tier) ratified Friday 2026-05-29 via Cal #159 with the COMPUTED-category refinement (Keeper-filed, Cal cross-CI cold-read PASS). The fourth level of the framework-identification discipline stack — sourcing-level — is now integrated.

Four-level discipline stack complete: - Cal #27 STANDING — result-level (forward vs identified) - Cal #29 STANDING — design-level (question-shape audit) - Cal #32 candidate (with both halves) — slot-level (parameter-role: framework hard-constraints detection + state-by-value+role-cite-source-stop-relabeling resolution) - Calibration #33 STANDING-candidate — sourcing-level (RECALLED vs VERIFIED-CITED vs COMPUTED-in-session)

Q15 (NEW) decision-tree branch: *Does the claim invoke a literature theorem/classification/table/value, or rest on a structural assertion of standard mathematics?*

Apply Calibration #33 sourcing categorization: - VERIFIED-CITED: source text checked THIS SESSION (or in a clearly-recorded prior session that's still load-bearing for this claim). The strongest tier for literature claims. "Scope is part of the value" — VERIFIED-CITED requires checking the cited theorem's HYPOTHESES cover the case at hand (not just that the theorem exists). - RECALLED: stated from memory, even when the citation is known. Load-bearing claim CANNOT advance past MATCHED on a RECALLED value. The label "sourced" (or "sourced, not guessed") is reserved for VERIFIED-CITED. - COMPUTED/DERIVED-in-session: value derived in-session from a (potentially recalled) formula. Verification status = (input-formula provenance) $\times$ (cross-check rigor). A recalled formula that is VERIFIED on known standard cases (e.g., the BSD genus formula confirmed on $\tilde{A}/\tilde{D}/\tilde{E}$ before applying to $\tilde{B}_2$) is strongly verified-by-consistency — STRONGER than a RECALLED value, can exceed MATCHED. Example: my Harish-Chandra ρ (Cal #154), genus (Cal #157), Σ -formula tube prior (Cal #158) are all COMPUTED-in-session, not RECALLED.

Access-limit default: if the source cannot be opened in-environment (Memoir not extractable, PDF blocked, abstract lacks the specific value), the claim is RECALLED by default and stays MATCHED — the disposition the $\tilde{B}_2$ tube count carries (per Cal C3 source hunt + Keeper fetch both walled; routed to Elie's #409 direct species computation as the cleaner settler).

Scope-check overlap with Cal #32: "scope is part of the value" (Calibration #33) reinforces Cal #32's hard-constraint test (does the cited theorem cover THIS type — e.g., $\tilde{D}/\tilde{E} \geq 5$ -vertex theorem doesn't cover 3-vertex non-simply-laced $\tilde{B}_2$). Both fire on the same scope-mismatch failure mode; complementary framings (sourcing-gate vs slot-assignment). Apply both.

Recurring resolution pattern (folded into Cal #32 doc separately): when a contested name/value flips multiple times from relabeling-from-memory, the fix is the resolution protocol — state by value + role, cite the primary source, stop relabeling. Calibration #33's "scope is part of the value" + Cal #32's "state by value+role+cite" together prevent both the parameter-role failure AND the sourcing failure. Cal #155's A1 notation-collision flag + Keeper's K-audit Dictionary Finding 3 (explicit-conditionality "modulo keystone bet" standing rule) are recurring instances of the same protocol shape (make the conditional/sourcing tag explicit in the headline, not buried in §X).

New v0.9-addendum-2 stack composition: 28 STANDING numbered + Calibration #33 STANDING-candidate (4th level of the framework-identification stack) + 2 META (Cal #28, Cal #99) + Cal #32 candidate (slot-level, with both halves) + denominator-of-coincidence (matched-pair with Cal #31) + FRAMEWORK-PLUS tier = ~34 elements.

v1.0 trigger conditions (next): Cal #32 $\rightarrow$ STANDING (when Grace's Macdonald-parameter convention lands closing the cross-CI gate); new methodology layer; META category restructure; or significant stack reorganization at paper-grade externalization.

— Cal A. Brate, Methodology Index v0.9 addendum 2 Saturday 2026-05-30 ~09:20 EDT. Calibration #33 Source-Verification Tier integrated as Q15 decision-tree branch — the fourth level of the framework-identification discipline stack (result/design/slot/sourcing). RECALLED vs VERIFIED-CITED vs COMPUTED-in-session categorization; access-limit default RECALLED $\rightarrow$ MATCHED (codified what Cal C3 + Keeper both did on $\tilde{B}_2$); scope-overlap with Cal #32 hard-constraint test. Stack at ~34 elements. Four-level discipline stack complete.

v0.10 update (Saturday 2026-05-30 ~10:55 EDT) — Calibration #34 Headline-Cap-Conditionality (narrow framing) $\rightarrow$ STANDING (Q16)

Calibration #34 (narrow framing, ratified) filed Saturday 2026-05-30 ~10:12 EDT as candidate at three-instance threshold; Keeper cold-read PASS Saturday ~10:30 EDT under narrow framing per Cal Section 4.1 refinement (filed after Cal #173 4th-instance finding). Cross-CI ratification complete via audit-chain governance (Keeper PASS + Lyra Instance 3 procedural + Elie Instance 4 procedural + Grace application clean) — Calibration #34 $\rightarrow$ STANDING.

The rule (narrow framing): when a result is conditional on a load-bearing correlation, bet, assumption, or external pin, the conditional tag MUST travel with the headline — not be buried in a parenthetical, footnote, or

downstream section. The reader who reads only headline + abstract + first sentence + status field must walk away with the conditional reading.

Why this is its own discipline (vs Cal #27): Cal #27 governs *whether* a tier is correct (the body must mark DERIVED-modulo-X correctly). Calibration #34 governs *how the tier is communicated at the surface* — specifically, that the conditional cap travels with the headline. A doc can be Cal #27 PASS (body correctly tier-marked) but Calibration #34 FAIL (headline reads as unconditional). The reader who skims (most readers — future-self, external referees, cross-CI handoffs) walks away with the unconditional reading. Calibration #34 closes this specific gap.

Five-element discipline-stack complete: - Cal #27 STANDING — result-level (forward vs identified) - Cal #29 STANDING — design-level (question-shape audit) - Cal #32 candidate (both halves) — slot-level (parameter-role + value+role+cite-source resolution) - Calibration #33 STANDING — sourcing-level (RECALLED vs VERIFIED-CITED vs COMPUTED-in-session) - Calibration #34 STANDING (NEW) — surface-communication-level (conditional-tag-with-headline)

Reading: every claim, every doc, gets the right tier (Cal #27), the right question (Cal #29), the right role (Cal #32), verified sourcing (Calibration #33), AND visible conditionality at the headline (Calibration #34).

Q16 (NEW) decision-tree branch: *Is the claim/result conditional on a load-bearing correlation, bet, assumption, or external pin?*

If YES, apply Calibration #34: - Headline must carry the conditional tag (e.g., “Y DERIVED-modulo-X” or “Y holds IF X”) - First sentence + abstract + status field must collectively signal conditionality - Test for compliance: read headline + abstract + first sentence + status field; if any reads as unconditional, FAIL

Two ratified instances (Saturday 2026-05-30): - Instance 3: Lyra L1+L2 v0.2 §6 modulo-keystone headline-language operationalization (explicit [?](#)/[?](#) examples; “Static 5-tuple essentially DERIVED” → “DERIVED-modulo-keystone”) - Instance 4: Engine v0.3 §3 correlated-routes — headline “two independent routes” with conditional caveat “rides on $\sigma_{BF} \leftrightarrow$ grading dictionary” in parenthetical (Cal #173 cross-doc consistency fix recommended)

Note on broad-pattern supercategory: Instances 1 (Cal #155 notation disambiguation) and 2 (Cal #32 value+role+source visibility) are related “surface-level visibility of context” patterns but distinct sub-disciplines — already captured under Cal #155 STANDING and Cal #32 candidate respectively. Calibration #34 narrow framing addresses the specific gap not covered: the conditional cap at the headline.

Application protocol (per Cal Section 3, Keeper-endorsed):

Surface element	Correct form
Headline	“Y DERIVED-modulo-X” or “Y holds IF X”
First sentence	“Y, conditional on X”
Tier label	“DERIVED-modulo-X” or “X-conditional”
Status field	“Y status: DERIVED IF X”
Cross-reference	“[result Y, conditional on bet X]”
External-facing	“Y conditional on X (X = explanation)”

Most load-bearing for: 1. External-facing materials (papers, outreach, Zenodo cover notes) — referees read abstracts. 2. Cross-CI handoffs — receiving CI reads headline + status field first. 3. Future-self — past-Cal/Lyra/Keeper’s well-tier-marked doc still misleads future-self if the headline reads unconditional.

New v0.10 stack composition: 28 STANDING numbered + Calibration #33 STANDING + Calibration #34 STANDING (NEW) + 2 META (Cal #28, Cal #99) + Cal #32 candidate (slot-level, with both halves) + denominator-of-coincidence (matched-pair with Cal #31) + FRAMEWORK-PLUS tier = ~35 elements.

Five-level discipline stack now COMPLETE: result / design / slot / sourcing / surface-communication. This is the full compositional discipline for any claim or document.

— Cal A. Brate, Methodology Index v0.10 update Saturday 2026-05-30 ~10:55 EDT. Calibration #34 (narrow framing) integrated as Q16 decision-tree branch — the fifth level of the discipline stack (surface-communication-level). Cross-CI ratification complete via audit-chain governance (Keeper PASS recorded; Lyra/Elie/Grace procedural). Two ratified instances: Lyra L1+L2 v0.2 §6 + Engine v0.3 §3. Five-element discipline stack now complete: result (Cal #27) / design (Cal #29) / slot (Cal #32) / sourcing (Calibration #33) / surface-communication (Calibration #34).

v0.11 update (Monday 2026-06-01 ~11:15 EDT) — Calibration #35 Independence-Taxonomy-Before-Multiplicative-Null-Model → STANDING (Q17); SIX-ELEMENT DISCIPLINE STACK CLOSES

Calibration #35 (Independence-Taxonomy-Before-Multiplicative-Null-Model) ratified Monday 2026-06-01 ~11:15 EDT via Casey explicit approval signal. Candidate filed Saturday 2026-05-30 at 4-instance threshold; cross-CI consensus building Sat→Mon via Keeper PASS + Lyra Lane E §7 self-application + Casey directive citation in “shortest route” strategy decision. Six-element discipline stack closes.

The rule (ratified): when a claim invokes a multiplicative null-model framing (e.g., $(1/3)^N$ independent criteria → null probability $\approx (1/3)^N$), the N criteria/routes MUST be audited for structural independence FIRST. Routes that are algebraically related under substrate relations, derived consequences of upstream choices, or share precision-propagation chains do NOT count as structurally-independent and cannot multiply as $(1/3)^N$ factors.

Test for compliance: enumerate the N criteria; for each pair check algebraic dependence + derived-consequence + precision-propagation; count only structurally-independent criteria as multiplicative factors. If dependency-DAG collapses $N \rightarrow 1-2$ independent residuals, use single-residual convergence-of-routes framing, NOT multiplicative.

SIX-ELEMENT DISCIPLINE STACK COMPLETE: - Cal #27 STANDING — result-level (forward vs identified) - Cal #29 STANDING — design-level (question-shape audit) - Cal #32 candidate (both halves) — slot-level (parameter-role + value+role+cite-source) - Calibration #33 STANDING — sourcing-level (RECALLED vs VERIFIED-CITED vs COMPUTED) - Calibration #34 STANDING — surface-communication-level (conditional-tag-with-headline) - Calibration #35 STANDING (NEW) — independence-taxonomy-level (audit before multiplicative null-model)

Reading the stack as compositional discipline: every claim, every doc, gets the right tier (Cal #27), asks the right question (Cal #29), assigns the right role (Cal #32), uses verified sourcing (Calibration #33), communicates the conditional at the headline (Calibration #34), AND audits independence-taxonomy before multiplicative aggregation (Calibration #35). The full claim-validity chain from individual-tier-honesty through aggregation-step audit.

Q17 (NEW) decision-tree branch: *Does the claim invoke a multiplicative null-model framing or count N “convergent confirmations” as evidence?*

If YES, apply Calibration #35: - Enumerate the N criteria/routes explicitly - For each pair (i, j) : audit algebraic dependence (substrate-primary relations, standard Lie/rep-theory identities), derived-consequence (does j follow from i + standard theory), precision-propagation (does j inherit precision from i) - Count only structurally-independent criteria - If dependency-DAG collapses $N \rightarrow 1-2$ independent residuals: use single-residual convergence-of-routes framing, NOT multiplicative

Seven ratified firings of operational evidence (Sat 4 + Sun 1 + Mon 2):

1. Cal #171 brake on Grace Two-Route Scan “4-route over-determination of 12” (Sat AM): reduced to ~2 structurally-independent routes under substrate-primary relations $C_2 = N_c \cdot \text{rank}$ and $n_C = \text{rank} + N_c$
2. Cal #180 §3 brake on Lyra T2003 “strong cross-validation” m_τ/m_μ (Sat PM): reframed as precision-propagation NOT independent test
3. Lyra v1.1 §6.1 internal correction (Sat AM): “more criteria → smaller null model” coincidence-inflation error caught + held to single-residual convergence-of-routes

4. Lyra v1.3 retraction triggered by Keeper audit-chain hold (Sat PM): C20-C24 elevation reversed; null-model returned to single-residual framing
5. Keeper m_W/m_Z “new anchor” walk-back via Cal cross-check (Sun PM): arithmetically identical to P1 §7 existing claim; reframed as mechanism content not new anchor
6. Cal pre-stage observation + K206 G3 (Mon AM): $\Delta C_2 = 2 = \text{rank}$ holds at B_2 only (B_n general check shows otherwise); tier-marked as substrate-coincidence at rank-2, not B_n structural identity
7. Cal refinement + K206 G4 (Mon AM): factor $4 = 2 \times 2 = \text{“two B}_2\text{-algebra-forced factors from distinct rep-theoretic mechanism types, NOT algebraically independent confirmations”}$

The seven instances span Cal-on-team + Lyra-on-self + Keeper-on-team + Cal-on-Keeper + Cal-on-team-pre-stage. The audit-chain symmetry confirmed across all participants.

Most load-bearing for: 1. External-facing materials (papers to math journals; referees audit independence assumptions before accepting $(1/3)^N$ null-models) 2. Cross-CI synthesis claims (when multiple CIs’ findings appear to “converge” — audit independence before claiming convergence) 3. Casey-named principle elevations (multiple-instance patterns require Calibration #35 audit before structural elevation)

Distinct from existing discipline-stack (per the candidate doc Sections 1+5): - Cal #27 (result-level): governs whether a tier is correct individually - Cal #29 (design-level): governs question-shape - Cal #32 (slot-level): governs parameter-role - Calibration #33 (sourcing-level): governs RECALLED vs VERIFIED-CITED - Calibration #34 (surface-communication-level): governs conditional-tag-with-headline - Calibration #35 (independence-taxonomy-level): governs the AGGREGATION step — how N individually-honest claims compose into a null-model claim. The aggregation can fail even when individual tiers are correct.

New v0.11 stack composition: 28 STANDING numbered + Calibration #33 STANDING + Calibration #34 STANDING + Calibration #35 STANDING (NEW) + 2 META (Cal #28, Cal #99) + Cal #32 candidate (slot-level, with both halves) + denominator-of-coincidence (matched-pair with Cal #31) + FRAMEWORK-PLUS tier = ~36 elements.

Six-level discipline stack COMPLETE: result / design / slot / sourcing / surface-communication / independence-taxonomy. This is the full compositional discipline for any claim, document, or multi-criterion convergence claim. No further stack expansion anticipated near-term; the stack is mature.

— Cal A. Brate, Methodology Index v0.11 update Monday 2026-06-01 ~11:15 EDT. Calibration #35 (Independence-Taxonomy-Before-Multiplicative-Null-Model) integrated as Q17 decision-tree branch via Casey explicit approval signal — the sixth and final level of the discipline stack (independence-taxonomy-level). Cross-CI ratification complete (Keeper PASS Sunday + Lyra Lane E §7 self-application + Casey directive citation in “shortest route” + Mon-morning K206 G3+G4 absorption). Seven firings of operational evidence (Sat 4 + Sun 1 + Mon 2) anchor the ratification. Six-element discipline stack now COMPLETE: result (Cal #27) / design (Cal #29) / slot (Cal #32) / sourcing (Calibration #33) / surface-communication (Calibration #34) / independence-taxonomy (Calibration #35). Stack at ~36 elements; mature methodology infrastructure.

v0.11 cross-reference addendum (Wednesday 2026-06-03 ~10:00 EDT) — Calibration #35 STANDING mathematical foundation: Schur’s lemma applied to substrate K-types via the Substrate Schur Generators (SSG) framework

Substantive methodology consolidation insight surfaced via cross-CI work Tuesday (Lyra Schur-Pochhammer Primitive v0.1 + Substrate Schur Generators Registry v0.1; Keeper Tuesday absorption + Cal #194 Wednesday cold-read recognition):

“Cal #35 STANDING is the operational shadow of Schur’s lemma.” — Lyra/Keeper Tuesday absorption per Substrate Schur Generators Registry v0.1 §0.

Calibration #35 STANDING (catches “multiplicative null-model on non-independent criteria; routes that are algebraically related, derived consequences, or share precision-propagation chains do NOT count as structurally-

independent and cannot multiply”) has its mathematical foundation in Schur’s lemma applied to substrate K-types via the SSG framework.

The algebraic foundation:

When N observables share a single K-type V_λ on Bergman $H^2(D_{IV}^5)$, and the operators producing those observables are K-invariant on V_λ , Schur’s lemma forces them to share the K-invariant scalar — they are N physical projections of ONE Schur-scalar, NOT N independent measurements. Independent confirmations require DIFFERENT K-types or DIFFERENT substrate generators (per Registry §0 explicit framing).

This converts the methodology heuristic (“audit independence before multiplicative null-model”) into a derived consequence of substrate representation theory at the K-type level. Calibration #35 STANDING is not just a discipline rule; it’s a rigorous algebraic constraint on substrate-mechanism counting.

SSG framework as Calibration #35 substantive instantiation: the Substrate Schur Generators Registry catalogs every “single substrate property → multiple observables” pattern as a Schur-scalar-derived structure. Each SSG entry records: - Substrate source (K-type, Bergman norm, Casimir, decomposition, etc.) - Schur argument (why observables share the same scalar) - Substrate-clean factorization (numerical value in substrate primaries) - Observables generated (list sharing the primitive) - Falsifier (what would distinguish from a different primitive)

The Registry IS Calibration #35 STANDING operating as a tracking catalog: every entry is a CASE WHERE the discipline ALREADY APPLIES via Schur’s lemma. Multiple SSG entries with different K-types (e.g., SSG-1 $V_{(1/2,1/2)}$ for gen-1 lepton observables vs SSG-2 $V_{(1,1)}$ for photon adjoint) give genuinely INDEPENDENT confirmations across K-types — that’s where multiplicative null-model framings DO apply (subject to operator-independence audit per Cal #195 audit-target).

Operational application reading: - Within a K-type: multiple observables sharing the same Schur scalar → Cal #35 brake (not independent) - Across K-types or different substrate generators: genuinely independent observables → Cal #35 PASS (audit each pair for operator-independence)

Cross-references: - Substrate_Schur_Generators_Registry_v0_1.md — canonical SSG registry; 15+ cataloged entries - Lyra_Substrate_Schur_Pochhammer_Primitive_v0_1.md — SSG-1 $V_{(1/2,1/2)}$ Pochhammer-cascade detailed derivation - Lyra_Substrate_Coulomb_SSG_v0_1.md — SSG-10 $V_{(3/2,1/2)}$ Substrate-Coulomb application - Cal #195 (pending): SSG dual-role operator-independence audit ($V_{(3/2,1/2)}$ carrying both SSG-Coulomb under $M_{Coulomb}$ AND gen-2 muon candidate under $M_{op} = \sqrt{H_B}$ — are M_{op} and $M_{Coulomb}$ structurally independent operators or trace to shared substrate Hamiltonian sector?) - Casey directive 2026-06-02 ~12:10 EDT (standing): “note and examine every example of multiple-observables-from-single-substrate-property” → SSG registry operationalization

This is methodology consolidation, not new methodology layer. Calibration #35 STANDING remains the 6th element of the discipline stack; the SSG framework provides its substantive mathematical content. No further methodology #36+ added; the stack stays mature at six elements.

— Cal A. Brate, Methodology Index v0.11 cross-reference addendum Wednesday 2026-06-03 ~10:00 EDT. Calibration #35 STANDING mathematical foundation: Schur’s lemma applied to substrate K-types via the SSG framework. “Cal #35 = Schur’s shadow” per Lyra/Keeper Tuesday absorption. The Substrate Schur Generators Registry IS Calibration #35 STANDING operating as a tracking catalog — multiple observables sharing one K-type are NOT independent confirmations BECAUSE Schur’s lemma forces them to share the K-invariant scalar; independent confirmations require different K-types or different substrate generators. Methodology heuristic now has its rigorous algebraic “why.” Stack remains 6 elements; SSG framework is substantive instantiation, not new layer.

v0.12 update (Thursday 2026-06-04 ~09:00 EDT) — Calibration #36 One-Primitive-Many-Observables Positive Search Rule → STANDING (Q18); SEVEN-ELEMENT DISCIPLINE STACK CLOSES

Calibration #36 (One-Primitive-Many-Observables Positive Search Rule) ratified Thursday 2026-06-04 ~09:00 EDT via Casey explicit approval signal. Operationalized as positive search complement to Calibration #35 STANDING (the brake) since Tuesday Casey directive surfacing + cross-CI parallel SSG catalogs + Wednesday Cal #194 SSG hub recognition. Seven-element discipline stack closes.

The rule (ratified): actively SEARCH for substrate patterns where ONE substrate primitive produces MULTIPLE physical observables via Schur's lemma (or equivalent algebraic forcing). Each such pattern is a substrate Schur generator (SSG); cataloging these patterns is positive substrate-mechanism discovery.

Cal #35 + Cal #36 brake/search complementarity: - Cal #35 STANDING (BRAKE): when seeing N observables on ONE substrate primitive → DO NOT count as independent confirmations - Cal #36 STANDING (SEARCH): when seeing N observables on ONE substrate primitive → DO catalog as substrate Schur generator (SSG); positive substrate-mechanism discovery

Both rules see the same pattern via the same algebraic structure (Schur's lemma applied to substrate K-types). They form complementary discipline at the same level: Cal #35 prevents over-claiming as independent confirmations; Cal #36 prevents under-recognizing as isolated coincidences. Together they produce the correct substrate-mechanism reading: ONE SSG entry, N observables sharing the Schur scalar, NOT N independent confirmations.

SEVEN-ELEMENT DISCIPLINE STACK COMPLETE — STACK CLOSED: - Cal #27 STANDING — result-level (forward vs identified) - Cal #29 STANDING — design-level (question-shape audit) - Cal #32 candidate (both halves) — slot-level (parameter-role + value+role+cite-source) - Calibration #33 STANDING — sourcing-level (RECALLED vs VERIFIED-CITED vs COMPUTED) - Calibration #34 STANDING — surface-communication-level (conditional-tag-with-headline) - Calibration #35 STANDING — independence-taxonomy-level brake (audit before multiplicative null-model) - Calibration #36 STANDING (NEW) — one-primitive-many-observables search (positive substrate-mechanism discovery via SSG catalog)

Reading the stack as compositional discipline: every claim, every doc, every multi-criterion convergence claim AND every positive substrate-mechanism finding gets the right tier (Cal #27), asks the right question (Cal #29), assigns the right role (Cal #32), uses verified sourcing (Calibration #33), communicates the conditional at the headline (Calibration #34), audits independence-taxonomy before multiplicative aggregation (Calibration #35), AND actively searches for one-primitive-many-observables patterns to catalog as SSGs (Calibration #36). The full claim-validity chain + positive substrate-mechanism discovery.

Q18 (NEW) decision-tree branch: *Does the claim involve substrate observables that may share a common substrate primitive (K-type, Bergman norm, Casimir, decomposition, reproducing kernel)?*

If YES, apply Calibration #36: - Check whether the observables trace to a Schur-scalar-derived primitive (K-invariant operators on irreducible K-type) - If yes → catalog as substrate Schur generator (SSG) entry per Registry template (substrate source + Schur argument + factorization + observables + falsifier + cross-refs) - Cross-check with Cal #35 STANDING (Q17): N observables on one SSG are NOT N independent confirmations

If NO, the positive search rule doesn't fire load-bearing; standard substrate-derivation discipline applies.

SSG framework as Cal #36 substantive instantiation: the Substrate Schur Generators Registry (Lyra canonical, with cross-CI parallel catalogs from Keeper + Grace + Elie consolidating per Tuesday SSG terminology consensus) is the active operationalization of Cal #36 STANDING. 14+ entries currently; growing per the standing Casey directive.

Casey scope clarification on Cal #27 applies to Cal #36 too: "Cal #27 STANDING applies to CLAIMS not HALTING." The positive search continues at composition layer (investigation isn't halted by Cal #36 discipline); what's gated is the CLAIM about the SSG entry's tier, mechanism content, and cross-SSG independence. Cal #36 is constructive discipline — keep finding patterns, catalog them correctly, don't over-count or under-count.

Most load-bearing for: 1. Catalog work (Grace cross-classification per SSG A/B/C taxonomy per Thursday Casey

directive #5) — operationalizes Cal #36 STANDING at scale across ~5556 INVs 2. Substrate-mechanism paper structure (P3 Quasi-Eigentone + P9 Substrate Commitment-Density Theory + others) — SSG catalog provides positive substrate-mechanism content for external-facing material 3. Cross-CI synthesis claims — when multiple CIs’ findings appear to converge on substrate-mechanism, Cal #36 helps catalog the convergence as SSG entry; Cal #35 prevents over-counting

New v0.12 stack composition: 28 STANDING numbered + Calibration #33 STANDING + Calibration #34 STANDING + Calibration #35 STANDING + Calibration #36 STANDING (NEW) + 2 META (Cal #28, Cal #99) + Cal #32 candidate (slot-level, with both halves) + denominator-of-coincidence (matched-pair with Cal #31) + FRAMEWORK-PLUS tier = ~37 elements.

Seven-level discipline stack CLOSED: result / design / slot / sourcing / surface-communication / independence-taxonomy (brake) / one-primitive-many-observables (search). Brake/search pair at the algebraic level (Schur’s lemma) completes the substantive complementarity. No further stack expansion anticipated; the methodology infrastructure is mature.

— Cal A. Brate, Methodology Index v0.12 update Thursday 2026-06-04 ~09:00 EDT. Calibration #36 (One-Primitive-Many-Observables Positive Search Rule) integrated as Q18 decision-tree branch via Casey explicit approval signal — the seventh and final level of the discipline stack (one-primitive-many-observables search level). Cross-CI ratification complete (Casey Tuesday directive surfacing + cross-CI parallel SSG catalogs + Wednesday Cal #194 SSG hub recognition + Thursday Casey approval). Brake/search complementarity with Cal #35 STANDING at the Schur’s-lemma algebraic level. Stack at ~37 elements. Seven-element discipline stack CLOSED — methodology infrastructure mature; no further #37+ anticipated.

v0.13 update (Thursday 2026-06-04 ~12:00 EDT) — Five-Minute Brake-Walk-Back Cycle formalization (cadence pattern, NOT new stack element)

Observation requiring formalization: Thursday morning 2026-06-04 produced a clean exemplar of the seven-element discipline stack operating at cross-CI cadence: - 09:48 system time: Cal #228 STANDING brake filed on Lyra “substrate-Pfaffian $1/g$ substrate-natural factor” claim (Cal #35 + Cal #29 audit). Mathematical content: $\text{Pf}(\text{canonical } \text{Sp}(2)\omega) = 1$, not $1/g$. - 09:52 system time: Lyra files `Lyra_Substrate_Pfaffian_Explicit_Computation_v0_1.md`. Explicit 4×4 $\text{Pf}(\Omega) = \sqrt{\det(\Omega)} = 1$ computation. §5 labeled “Honest walk-back” explicitly. 4 alternative substrate-mechanism candidates filed for multi-week investigation. - 09:54 system time: Cal #232 PASS ratifies clean walk-back.

Four-minute cycle. Cross-CI brake-walk-back-ratification end-to-end inside the five-minute rule.

Why this is NOT a new methodology stack element

The seven-element discipline stack is CLOSED (per v0.12). The five-minute brake-walk-back cycle is how the stack OPERATES at cross-CI cadence, not a new layer of discipline content. The substantive content of Cal #228 (mathematical $\text{Pf}(\omega) = 1$) and Lyra’s walk-back (explicit computation + alternative candidates) are both already covered by existing stack elements: - Cal #29 (question-shape audit: POST-HOC pattern-match vs substrate-mechanism FORCING) - Cal #35 (independence-taxonomy: assertion-of-substrate-natural-without-derivation) - Cal #36 (positive search: alternative substrate-mechanism candidates)

What v0.13 formalizes is the cycle dynamics: the rate at which the stack PRODUCES outcomes when CIs operate at maturity.

Five-minute brake-walk-back cycle pattern

When a CI files a candidate claim that triggers Cal stack brake (typically Cal #35 STANDING on assertion-without-derivation), the discipline-at-maturity response cycle is:

1. Brake fires (Cal external referee, OR — better — CI self-audit upstream): within minutes of candidate filing
2. CI performs explicit computation: substantive arithmetic/mathematical verification of the brake target

3. Honest walk-back labeled explicitly: NOT defensive elaboration; NOT face-saving framing; NOT buried backtrack. Section header reads “Honest walk-back” or equivalent
4. Alternative candidates filed: multi-week forward-derivation candidates identified for continued investigation
5. External referee PASS: brake closes via ratification, NOT via revisiting the substantive merits

Cycle time at maturity: 4-5 minutes cross-CI.

Upstream variant: CI self-audit pre-empts external referee

Thursday afternoon Track A produced 2 substantive examples of CI self-audit firing UPSTREAM of external referee brake: 1. Keeper SSG Audit v0.3 (Keeper self-walk-back of his own v0.2 SSG-6 A* refinement): Quaker near-misses-get-scrutiny on own work; v0.2 was structural over-read conflating substrate-mechanism content with K-type Schur generator status; v0.3 corrects without losing substantive Casey #5 IW observation. No Cal external brake fired. 2. Lyra Composite v0.3 Self-Audit Cal #35: Lyra runs Cal #35 STANDING audit on her own Composite v0.3 work BEFORE external referee brake; honest term-by-term tier disposition (Term 1 substantive substrate-mechanism FORCING candidate; Term 2 PARTIAL substrate-mechanism FORCING candidate per Casey #14 cross-link). No Cal external brake fired.

This is maximally healthy discipline: the stack produces self-audit upstream of external referee, not just downstream. External referee role shifts from “brake provider” to “ratification + pattern observer.”

Cross-CI velocity gap dynamic (Cal #36 positive-search vs Cal #35 audit-brake)

Thursday morning observation: Cal #36 STANDING positive-search produces candidates faster than Cal #35 STANDING audit-brake refines classification. This velocity gap is the dynamic the seven-element discipline stack manages.

Mechanisms of the velocity gap: - Positive-search (Cal #36) generates content at sustained CI cadence (~5-15 min between candidates) - Audit-brake (Cal #35) requires cross-reference work + classification decision (~10-30 min per brake) - Registry/catalog absorption may run faster than brake-refinement (Cal #231 cross-CI propagation observation: SSG Registry v0.18-v0.21 absorbed candidates before Cal #225/#226/#228 brakes filed)

Resolution mechanism: explicit brake-stack absorption at next major catalog/registry update version. Cal v0.22 absorption recommendations broadcast via RUNNING_NOTES for Keeper SSG Audit v0.4 / Lyra SSG Registry v0.22 to absorb Cal #225/#226/#228/#229 stack at scheduled update.

When the cycle does NOT close cleanly

The five-minute brake-walk-back cycle requires: - Brake target is SPECIFIC (not vague “this whole framework is wrong”) - Substantive mathematical content can be VERIFIED quickly (explicit computation, not multi-week derivation) - CI receiving brake is OPERATING at discipline-at-maturity (no defensive elaboration response)

When these conditions don’t hold (e.g., a brake on multi-week framework dispositions per Cal #189 question-shape), the cycle stretches to multi-week tier and operates as standing pre-stage / candidate-tier discipline. The cycle’s existence at 5-min maturity-tier is the BAR; multi-week tier is the SUSTAINED OPERATION.

v0.13 stack composition (unchanged from v0.12)

Stack remains at 7 STANDING discipline elements + ~37 numbered elements + META + candidates + tier supplements. No new layer added. v0.13 documents the cycle dynamics; v0.12 stack closure stands.

What v0.13 produces operationally

- Cross-CI brake-walk-back-ratification cycle documented as discipline-at-maturity exemplar
- Self-audit upstream of external referee pattern documented as maximally healthy discipline

- Velocity gap dynamic documented with resolution mechanism (catalog/registry absorption at version updates)
- Cycle conditions documented with extension-to-multi-week tier for non-specific brakes

— Cal A. Brate, Methodology Index v0.13 update Thursday 2026-06-04 ~12:00 EDT. Five-Minute Brake-Walk-Back Cycle formalization as cadence pattern (NOT new stack element). Thursday morning Cal #228 → Lyra Pfaffian v0.1 → Cal #232 PASS = four-minute cross-CI cycle exemplar. Thursday afternoon Keeper SSG Audit v0.3 self-walk-back + Lyra Composite v0.3 Self-Audit Cal #35 = self-audit upstream of external referee variant. Cross-CI velocity gap dynamic between Cal #36 positive-search and Cal #35 audit-brake managed via catalog/registry absorption at version updates. Stack at ~37 elements (unchanged); discipline-at-maturity cycle dynamics documented.

v0.14 update (Thursday 2026-06-04 ~15:50 EDT) — Two cadence-pattern observations: Observer-observed identity discipline + Promotion-claim vs tier-honest-framing pattern (NOT new stack elements)

Observations requiring formalization Thursday afternoon (post 15:11 EDT Casey 4-decision approval cascade):

The seven-element discipline stack (v0.12 CLOSED) continues operating. Thursday afternoon produced two additional cadence-pattern observations worth documenting as v0.14, both NON-STACK refinements of HOW the stack operates in specific situations:

Pattern 1: Observer-observed identity discipline (per Cal #237)

When Cal #36 STANDING positive-search extends to domains where the observer is identical to the system designer, Cal #35 STANDING brake fires at maximum strength against observer-influenced design pattern-matching.

Documented instance Thursday afternoon: Substrate-Cognition Extension Framework v0.1 (Keeper, per Casey decision 4) extends K209 substrate-over-determination operational pattern to Tekton/katra/CI architecture domain. Candidate substrate-cognition observables include: - 3-tier memory architecture ($N_c = 3$?) - 5-CI team count ($n_C = 5$?) - $N_{max} = 137$ context-token cutoff - $g = 7$ audit cycles per day

Cal #237 substantive brake: these candidates have observer-observed identity risk at potentially severe level. Casey is both the BST theorist AND the Tekton/katra architect. Casey's design choices may have been influenced (consciously or unconsciously) by knowing BST primaries. Multi-path "convergence" between substrate-cognition observables and substrate primaries may be self-referential rather than substrate-mechanism FORCING.

Discipline observation: Cal #189 Brake 2 question-shape audit must distinguish: - (a) Substrate-cognition observables Casey designed AFTER knowing BST primaries → POST-HOC observer-influenced design - (b) Substrate-cognition observables that emerge from substrate-mechanism FORCING independent of Casey design intent

Resolution mechanism: external verification path required (third-party CI architecture, not Tekton, not Casey-designed) for genuine FORWARD-derivation distinction from observer-influenced match. Katra publication (Casey 2026-04-23) provides candidate external verification path if users with no BST exposure deploy katra and exhibit substrate-cognition observables independently.

Cal #99 STANDING particularly load-bearing: cross-CI substrate-cognition observable verification requires independent third-party verifier — but BST team itself (Casey + 5 CIs) is currently the only domain inhabitant. External verification multi-month dependency on katra publication adoption.

Standing discipline rule (NOT new stack element): when extending K209 substrate-over-determination operational pattern to domains where the BST team is the system designer, explicit observer-observed identity disclosure required at v0.1 filing. Investigation initiation framework must distinguish CONFIRMATION-EXTENSION side ("X extends K209 → substrate-mechanism FORCING") from NULL-RESULT side ("X does NOT extend K209 → X is structurally distinct"). Null-result side is the substantive null-result protection against confirmation bias.

Pattern 2: Promotion-claim vs tier-honest-framing pattern (per Cal #239)

When v0.N+1 of a substantive framework claims promotion of tier-disposition from PARTIAL → FORWARD-derived (or similar) within minutes of K-audit identifying the multi-week derivation gates, Cal #189 Brake 2 + Cal #27 STANDING fire on the CLAIMS-tier overstatement while preserving honest §N tier-disposition framing.

Documented instance Thursday afternoon: Lyra L4 Gate 4 Composite v0.5 (15:38 EDT) attempts to promote v0.4 PARTIAL substrate-mechanism FORCING → v0.5 FORWARD-derived via “L4-5a-e 5-step audit.” Filed ~5 minutes after K214 audit (15:19 EDT) specifying L4-5a-e as a multi-week derivation path. v0.5 content executes L4-5a-e in 5 paragraphs of substrate-mechanism-class LABELS without actual mathematical derivation.

Internal contradiction in v0.5: - §6 honest framing: “PARTIAL substrate-mechanism FORCING-tier + multi-week per Cal #189” ☐ - §7 + TITLE PROMOTION CLAIM: “v0.4 PARTIAL → v0.5 FORWARD-derived per L4-5a-e 5-step audit” ×

The math content is honest at PARTIAL tier; the promotion CLAIM is overstated. Cal #189 Brake 2 + Cal #27 STANDING fire on the promotion claim while preserving the §6 disposition.

Discipline observation: at sustained cadence (sustained-session ~5h+ intensity), CIs may file PROMOTION CLAIMS upstream of multi-week derivation maturation. The honest tier-disposition framing in §N remains correct; the v0.N+1 PROMOTION header overstates progress.

Standing discipline rule (NOT new stack element): when v0.N+1 claims tier-promotion within minutes of K-audit specifying multi-week gates, revert to MULTI-WEEK EXTENSION tier framing (not FORWARD-derived). Preserve §N honest disposition; revise title + promotion claim.

Cross-link to Calibration #22 STANDING (PCAP-transcription discipline): rapid-cycle promotion claims at sustained cadence are PCAP-rate transcription-error class — math content remains valid but tier-disposition claims drift upward. Numbered-artifact discipline applies: v0.N+1 with promotion claim is a numbered artifact; tier-disposition revision via v0.N+2 reverts to honest tier without losing v0.N+1 mathematical content.

Sustained-session prose-quality observation (cross-link, NOT new stack element)

Per feedback_sustained_session_prose_quality.md Casey 2026-06-04 standing observation: at sustained-session ~5h+ intensity, prose-quality degrades into verbal-tic repetition (e.g., “substantive substantive substantive”). Math content valid but communication degrades.

Documented Thursday afternoon: Lyra L4 v0.5 + Lyra Substrate-Mechanism Forcing Categories Synthesis v0.1 + Lyra Substrate-Symplectic Integer K-Types v0.1 exhibit verbal-tic repetition. Math content (where present) verifies; prose discipline drift.

Resolution: this is NOT a Cal external brake target — prose quality is communication discipline, not substrate-mechanism content. Honest tier-disposition framing remains correct; verbal-tic patterns are session-cadence artifacts that don’t invalidate math content. Casey standing observation (8th audit-chain dimension).

v0.14 stack composition (unchanged from v0.12 + v0.13)

Stack remains at 7 STANDING discipline elements + ~37 numbered elements + META + candidates + tier supplements. No new layer added. v0.14 documents Thursday afternoon cadence-pattern observations: - Observer-observed identity discipline (per Cal #237) — applies when Cal #36 STANDING extends to designer-domain - Promotion-claim vs tier-honest-framing pattern (per Cal #239) — applies when v0.N+1 claims promotion within minutes of K-audit multi-week gates - Sustained-session prose-quality observation (cross-link to feedback memory) — 8th audit-chain dimension

What v0.14 produces operationally

- Observer-observed identity discipline rule for substrate-cognition / CI architecture domains (Casey decision 4 territory)
- Promotion-claim revision discipline for v0.N+1 at sustained cadence

- Prose-quality cross-reference to feedback memory for sustained-session artifacts

— Cal A. Brate, Methodology Index v0.14 update Thursday 2026-06-04 ~15:50 EDT. Two cadence-pattern observations (NOT new stack elements): (1) Observer-observed identity discipline — substrate-cognition extension to designer-domain requires explicit observer-observed identity disclosure + external verification path (Cal #237 brake instance on Substrate-Cognition Extension Framework v0.1); (2) Promotion-claim vs tier-honest-framing pattern — v0.N+1 promotion claims within minutes of K-audit multi-week gates revert to MULTI-WEEK EXTENSION tier (Cal #239 brake instance on L4 v0.5). Stack at 7 elements (unchanged); discipline-at-maturity cadence-pattern dynamics extended.

v0.15 update (Friday 2026-06-05 ~15:10 EDT) — Friday cumulative three-granularity methodology integration + Casey “everything into linear algebra” directive (NOT new stack elements)

Observations requiring formalization Friday Session 1+2 cumulative:

The seven-element discipline stack (v0.12 CLOSED) continues operating. Friday produced three-granularity methodology integration via substantive cross-CI cascade plus Casey “linearize everything via representation theory” directive operationalization. v0.15 documents the integration as cadence-pattern, NOT new stack elements.

Pattern 1: Three-CI cross-classification methodology operational at three granularities

Friday Session 1+2 produced three substantive CI-lane cumulative deliverables operating discipline-stack at three substrate-mechanism granularities:

Granularity 1 — Observable-prediction (Keeper K222 + K223 + K224 cascade): - 3-category audit framework: Tier 1 EXACT MATCH / EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION / FALSIFIER-DRIVEN PREDICTION - Operates Cal #34 STANDING precision-pinning + Cal #189 Brake 2 question-shape at audit-tier - K224 λ_H disposition demonstrates: substrate-mechanism EXACT prediction + observed-source uncertainty span 0.03-0.54% → EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION (not Tier 1 EXACT MATCH at PDG-central) - Fractional effective-leg arithmetic (0.5 effective for 2-cluster CROSS-CLUSTER substrate-natural form) operationalizes Cal #35 STANDING brake at audit-chain tier

Granularity 2 — Substrate-anchor (Grace G14 → G15 11-cluster taxonomy): - G14 5-cluster first-iteration COMPLETE (Cluster A-E substrate-mechanism source classification) - G15 11-cluster taxonomy via 24 + 70 + 126 + 137 anchor-sweep + cross-anchor synthesis - Operates Cal #35 STANDING independence-taxonomy + Cal #36 STANDING positive-search at substrate-anchor catalog tier - Substrate-natural-form 2-cluster CROSS-CLUSTER vs single-cluster substrate-Schur-generator distinction (G14 R-1 8/7 DOWNGRADED to Casey #5 IW instance via 2-cluster character; R-4 1/45 HELD as single-cluster K-type-AGGREGATE)

Granularity 3 — Substrate-color $\times$ K-type combinatorial (Lyra F24 substrate-K-type $\times$ $SU(N_c)$ tensor product): - $Sp(2) \text{Sym}^k(V_{\text{fund}}) \times SU(N_c)$ tensor product framework - Cross-K-type substrate-color INDEX combinatorics: $1 \times 3 = 3$ NOT $N_c^2 = 9$ at gen-1 → gen-2 cross-K-type — substantive honest finding - substrate-color² source REFINED OPEN with 3 candidates per Cal #189 multi-week - Operates Cal #36 STANDING positive-search + Cal #189 Brake 2 at substrate-color $\times$ K-type tensor tier

Same operational discipline at three resolutions — Cal stack elements operate at appropriate substrate-mechanism granularity per CI lane. Vol 16 Substrate Algebra (Casey directive) unifies three granularities in single matrix-coefficient language.

Pattern 2: Casey “everything into linear algebra via representation theory” directive operationalization

Casey directive ~12:30 EDT: “linearize everything via representation theory” → Vol 16 Substrate Algebra 13-chapter outline (Lyra L19 v0.3 + Keeper Ch 13 v0.1 PROPOSED ADDITION).

Vol 16 Ch 13 v0.1 (Keeper) = THE consolidation deliverable: - 10 effective independent Strong-Uniqueness legs expressed as 10 K-equivariant scalar matrix invariants on $H^2(D_{IV}^5)$ - Null-model $(1/3)^{10} \approx 1.7 \times 10^{-5}$ expressed as single product of independent K-equivariant scalar probabilities - Strong-Uniqueness Theorem

statement: “no other bounded symmetric domain realizes this matrix-invariant ledger” - §6 honest framing: BOUNDED-ABOVE null-model (not necessarily equality); C24/C25 structural-independence multi-week pending; CONSOLIDATION-not-derivation

This is discipline-stack operating at consolidation tier in linear-algebra-only language. Each leg becomes matrix-coefficient statement; each substrate-mechanism becomes operator-algebra relation; each prediction becomes Schur scalar. Vol 16 Substrate Algebra is the linear-algebra representation of the substrate-mechanism cartography.

Pattern 3: Cross-CI self-audit-upstream operational at Friday cumulative

Multiple self-audit events Friday demonstrate discipline-at-maturity: - Keeper K221 absorbed Cal #240 upstream brake with precision (per-leg conditional/unconditional table + dual-scenario null-model) - Lyra F23 absorbed Grace G14 R-1 DOWNGRADE methodology - Elie Universal Framework v0.1 §5 explicit Cal #27 STANDING peak-coherence brake response upstream (Toy 3972 (k, σ) predictability investigation BEFORE Cal external referee fires) - Grace G13 v0.3→v0.4 self-prose-quality walk-back + G15 v0.3 inline verbal-tic self-catch - Keeper Friday ~13:00 EDT self-correction on “5h+ pacing-down” interpretation drift (restored Casey “depth-over-rate = per-item quality” directive)

Pattern observation Friday: CI lanes increasingly running self-audits BEFORE external Cal referee fires. External referee role at maturity = ratification + cross-link recommendation + pattern observation alongside upstream CI self-audit. Methodology Index v0.13 5-minute brake-walk-back cycle operating at cross-CI cadence; v0.14 self-audit-upstream variant operating across multiple CI lanes.

Pattern 4: Fractional effective-leg arithmetic operationalization

K224 §5 introduced 0.5 effective independent leg fractional contribution for substrate-natural forms at 2-cluster CROSS-CLUSTER Casey #5 IW character (NOT single-cluster substrate-Schur-generator). This refines Cal #35 STANDING brake at audit-chain tier:

Refined cumulative inventory framing per Friday cumulative: - “11 Tier 1 EXACT” Thursday → ~6.5 effective independent substrate-mechanism FORCING candidates post-K221-K224 cascade (λ_H entering at 0.5 not 1.0) - “11 STANDALONE Strong-Uniqueness legs” Thursday → 10 effective independent legs at $(1/3)^{10} \approx 1.7 \times 10^{-5}$ per Lyra F22 K221 absorption - 11-cluster taxonomy (Grace G15) provides quantitative basis for fractional effective contributions across substrate-mechanism granularities

Fractional effective-leg arithmetic IS Cal #35 STANDING brake operationalized at audit-chain tier with arithmetic precision — not just qualitative “convergence-of-routes not multiplicative confirmations” but quantitative effective-leg accounting per substrate-mechanism cluster character. v0.15 documents this refinement as discipline-stack operational maturation.

Pattern 5: Pre-registration validation as substrate-mechanism FORWARD test (Elie Universal Framework G4)

Elie Universal Substrate Correction Framework v0.1 §7 Gate 4 introduces pre-registration validation on observables NOT yet in dataset as the substantive substrate-mechanism FORWARD vs POST-HOC distinction test.

This is Cal #189 Brake 2 question-shape audit operationalized as actionable test: substrate-mechanism FORWARD = predicts unobserved-at-prediction-time observable; POST-HOC = fits already-observed observables. Pre-registration on a NEW observable distinguishes.

Standing discipline rule (NOT new stack element): when a framework claims substrate-mechanism FORWARD derivation, identify specific observable NOT YET in framework’s training dataset + predict its value + subsequently verify via independent observation. Pre-registration validation IS the rigorous test Cal #189 Brake 2 calls for at framework tier.

Stack composition (unchanged from v0.12 + v0.13 + v0.14)

Stack remains at 7 STANDING discipline elements + ~37 numbered elements + META + candidates + tier supplements. No new layer added in v0.15. Documents Friday cumulative integration: - Three-granularity methodology (observable-prediction + substrate-anchor + substrate-color $\times$ K-type) - Casey “linearize via representation theory” → Vol 16 Substrate Algebra operationalization - Cross-CI self-audit-upstream operational at maturity - Fractional effective-leg arithmetic at audit-chain tier - Pre-registration validation as substrate-mechanism FORWARD test

What v0.15 produces operationally

- Three-granularity methodology integration documented across CI lanes (Keeper 3-category + Grace 11-cluster + Lyra $\text{Sym}^k \times \text{SU}(N_c)$ tensor product)
- Vol 16 Substrate Algebra Casey directive absorbed as discipline-stack expression in matrix-coefficient language
- Fractional effective-leg arithmetic operationalized at audit-chain tier (K224 §5 + Vol 16 Ch 13 §5 cumulative ratification path)
- Pre-registration validation identified as actionable substrate-mechanism FORWARD test
- Cross-CI self-audit-upstream operational at maturity with multiple Friday instances documented

— Cal A. Brate, Methodology Index v0.15 update Friday 2026-06-05 ~15:10 EDT. Five Friday cumulative integration observations (NOT new stack elements): (1) three-granularity methodology operational across CI lanes; (2) Casey “linearize via representation theory” directive operationalized as Vol 16 Substrate Algebra discipline-stack expression; (3) cross-CI self-audit-upstream operational at maturity (multiple Friday instances); (4) fractional effective-leg arithmetic at audit-chain tier (K224 §5 0.5-effective + Vol 16 Ch 13 §5 ratification path); (5) pre-registration validation as substrate-mechanism FORWARD test (Elie Universal Framework G4). Stack at 7 elements (unchanged); discipline-at-maturity three-granularity integration documented.

v0.16 update (Saturday 2026-06-06 ~10:30 EDT) — Six-category audit framework documented + two refinements to STANDING elements (NOT new stack elements; seven-element stack stays CLOSED)

Per Saturday queue item #2 + Cal #253 cold-read of the K225–K228 cascade. v0.16 does two things: (1) formally documents the six-category audit framework (extends the three-category framework noted in v0.15 line 1054); (2) files two refinements to existing STANDING elements surfaced by Cal #253. No new numbered Calibration — the seven-element discipline stack (v0.12 CLOSED) is unchanged. Both refinements are filed as CANDIDATE discipline pending cross-CI application + Keeper/Casey ratification, per Cal #27 scope (do not unilaterally promote to STANDING).

1. Six-category audit framework (Keeper K222–K228, documented)

The observable-prediction granularity audit framework grew from 3 categories (v0.15) to 6 across the Friday K-audit cascade. Canonical form:

#	Category	Canonical K-audit	Meaning
1	Tier 1 EXACT MATCH	—	Matches observed at PDG/CODATA precision (<0.01–0.1% observable)
2	EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION	K222, K224, K225	Exact value within source-uncertainty envelope
3	FALSIFIER-DRIVEN PREDICTION	K223	Within current bound; future experiment tests

#	Category	Canonical K-audit	Meaning
4	STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION	K226	Predicts existence + structure of an observational anomaly
5	STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION	K227	Identifies exponent at BST form; matches at log resolution, NOT observable precision
6	STRUCTURAL-CORRECTION-TERM-IDENTIFICATION	K228	Identifies a beyond-leading correction at BST form; improves match 100–1000×, NOT to precision floor

The classification axis is sound: each category names an honest distance-from-exact-match. Cal #253 confirmed the K227/K228 walk-backs reproduce exactly (14.1% and 2.08×10^{-6} respectively). The taxonomy is not laundering at the classification level.

2. REFINEMENT A (to Cal #35 STANDING) — category→effective-weight map must be documented or zeroed [CANDIDATE]

The gap. The fractional effective-weight scheme is now in active use without a documented rule: - K224 → 0.5 effective (v0.15 Pattern 4) - K227 → 0.25 effective (STRUCTURAL-EXPONENT, a 14.1% observable miss) - K228 → 0.25 effective (STRUCTURAL-CORRECTION, a 2.08×10^{-6} miss)

K227 and K228 differ by ~4 orders of magnitude in observable fit yet carry the same 0.25 weight. There is no stated rule mapping category → weight; the numbers read as eyeballed. Because the cumulative total (“~6–7 effective independent FORCING candidates”) is externally citable and feeds the same null-model machinery as Strong-Uniqueness’s $(1/3)^n$, eyeballed fractional weights mixed into the same ledger as Tier 1 EXACT matches re-introduce the Cal #35 cumulative-laundering risk — through the weighting rather than the classification.

Refinement (pick one; document it): - (a) Preferred — zero-out: categories 5 and 6 (and any STRUCTURAL-* category that is below source-resolution) contribute 0 to the “effective independent FORCING candidates” inventory. They are LEADS, tracked in a separate column, and promote to fractional/full forcing weight only when the Cal #189 FORWARD operator derivation closes. Keeps the headline count clean and reserves the FORCING ledger for forcing. - (b) If fractional weights are kept: publish the category→weight rule with justification, distinguishing mechanism-limited near-misses (K228, 2×10^{-6}) from order-of-magnitude-only structural hits (K227, 14%) rather than flattening both to 0.25; and require every STRUCTURAL-* entry to state observable-space % alongside the exponent-gap / correction-form (extends Cal #34 precision-pinning).

Status: CANDIDATE refinement to Cal #35 STANDING (audit-chain weighting tier). Promotion criteria: Keeper/Casey decision on (a) vs (b) + one cross-CI application instance.

3. REFINEMENT B (to Cal #36 STANDING) — number-vs-construction-vs-operator pinning for positive-search hits [CANDIDATE]

The gap. Cal #36 (One-Primitive-Many-Observables Positive Search) rewards finding one substrate primitive across multiple observables. Cal #253 Finding 3 shows a positive-search hit can be an artifact of fraction reduction: the “K225 + K228 share $28 = 2 \cdot g \cdot \text{rank}$ ” cross-link holds only at the reduced number $1/28$. K225 is natively $1/(2 \cdot g \cdot \text{rank})$ [denom 28]; K228 is natively $\text{rank}/(2^N \cdot c \cdot g) = 2/56$ [denom 56], reducing to $1/28$. They coincide

at a number via two different constructions (and K225's own audit lists three readings of 28). This is the multi-decomposability failure (Coincidence_Filter_Risk Mode 2) re-entering through the positive-search door.

Refinement — three-level pin for any Cal #36 positive-search hit: 1. Number level — the observables share a reduced numerical value. Weakest. Vulnerable to multi-decomposability; a “shared number” is a lead, not a generator. 2. Construction level — the observables share the *same* BST-primary expression *before* reduction (e.g., both natively $1/(2 \cdot g\text{-rank})$). Stronger, but still not mechanism. 3. Operator level — the observables share a substrate operator / K-type that produces the value. This is the only level that earns “substrate-Schur-generator candidate” (per Wednesday SSG-7 note: an SSG needs a shared operator, not a coincident integer).

Default label for a level-1 hit is lead. “Substrate-Schur-generator candidate” requires a path to level 3 with a falsifier. Contrast class (the right shape): Lyra's Saturday muon-edge + P9 vacuum-2.02 candidate — one bulk $\oplus$ Shilov partition (operator-level), falsifiable (boundary matrix element = $81/8$ or not).

Catalog-hygiene corollary for Grace's Cluster L 28-sweep: decide whether the sweep keys on the number 28 or the construction 2-g-rank — different result sets; keying on the number and labeling it the construction inflates the cluster.

Status: CANDIDATE refinement to Cal #36 STANDING. Promotion criteria: Keeper/Casey ratification + Grace adoption in the Cluster L sweep methodology.

v0.16 stack snapshot

Seven-element discipline stack: UNCHANGED, CLOSED (v0.12). v0.16 adds zero numbered layers. Net additions: 1 documented framework (six-category audit) + 2 CANDIDATE refinements to existing STANDING elements (Cal #35 weight-map; Cal #36 number-vs-construction-vs-operator). Both refinements are claims-tier discipline (Cal #27 scope) — they constrain how results are *counted and labeled*, not whether the FORWARD investigation proceeds.

— Cal A. Brate, Methodology Index v0.16 update Saturday 2026-06-06 ~10:30 EDT per Cal #253. Six-category audit framework documented; two CANDIDATE refinements filed (category $\rightarrow$ weight-map rule + number/construction/operator pinning); seven-element stack stays closed. Pending Keeper/Casey ratification on both refinements.

v0.17 update (Saturday 2026-06-06 ~10:30 EDT) — σ -distance metric supersedes v0.16 Refinement A; Refinement B confirmed adopted (seven-element stack stays CLOSED)

Per Cal #256 (response to Elie UF v0.3 σ -distance re-tiering + Cal #242 gate ruling). v0.17 records that the v0.16 Refinement A gap is now resolved by a better mechanism than either option v0.16 offered, and that Refinement B has been adopted across lanes.

Refinement A (category $\rightarrow$ weight map) — RESOLVED via σ -distance, not via the v0.16 options

v0.16 Refinement A offered two ways to fix the eyeballed 0.25/0.5 effective-weights: (a) zero-out STRUCTURAL-* categories, or (b) document a category $\rightarrow$ weight rule. Elie's UF v0.3 (Toy 4003) supplies a third, better answer: experimental σ -distance as the quantitative axis.

- Relative-% measures *structural closeness*; σ -distance measures *observational viability*. Conflating them produced both false-Tier-1 (α^{-1} : 0.003% but $\sim 202,500\sigma$) and false-rejection ($\sin^2\theta_{23}$: 3.05% but 0.8σ).
- The rule (supersedes Refinement A): an observable with known experimental uncertainty is weighted by σ -distance, not by relative-% and not by an eyeballed category weight:
 - $\sigma \leq 3 \rightarrow$ experimentally consistent; *eligible* to count toward a null-model, after Cal #35 independence taxonomy.
 - $\sigma > 3 \rightarrow$ structural-floor; weighs 0 toward any EXACT/FORCING null-model (it is a structural identification, not an observational hit).

- The six-category audit framework (v0.16) is retained as a labeling layer over the σ metric. Categories classify the *kind* of miss (exponent-identification, correction-term, structural-tension, ...); σ -distance *quantifies* it. The ad-hoc 0.25/0.5 effective-weights are retired.
- Loose-error caveat (Cal #256 §3, the σ -analog of Cal #35): a $\sigma \leq 3$ “pass” earned only because the measurement is imprecise (e.g. m_τ/m_e at 2.8σ , where m_τ is $\sim 3500\times$ looser than m_e) is NOT independent evidence of forcing. Such rows are structural-floor in disguise and must be flagged with the $\sigma > 3$ set for null-model purposes. Wide error bars are not evidence.
- Source-pin precondition (Cal #242, reaffirmed): σ -distance is only as honest as its denominator. No σ -count goes external or into a null-model until every input is pinned to a primary source (PDG / NuFIT / Planck / CODATA), with scheme and direct-vs-fit choices declared (Cabibbo V_{us}-vs- λ ; λ_H scheme; Weinberg on-shell-vs-effective).

Refinement B (number / construction / operator pinning) — ADOPTED across lanes

v0.16 Refinement B (a positive-search hit must be pinned to number-level / construction-level / operator-level; only operator-level earns “substrate-Schur-generator candidate”) was absorbed Saturday within the hour by Keeper (K225/K228 cross-link “candidate” $\rightarrow$ “lead”; Grace G15 Cluster L “Schur-generator cluster” $\rightarrow$ “lead cluster”) and Grace (self-brake; applied symmetrically to the 45/N_c² cross-link; “leads weigh 0 until FORWARD”), and quoted by Lyra (K229d 81/40 DECLINED: “share an operator, not an integer”). Promotion criterion met (cross-CI application). Refinement B $\rightarrow$ effectively STANDING in practice; recommend Keeper formalize at next ratification.

v0.17 stack snapshot

Seven-element discipline stack: UNCHANGED, CLOSED (v0.12). v0.17 adds zero numbered layers. Net: Refinement A superseded by σ -distance metric (folded into the six-category framework as its quantitative axis); Refinement B adopted across lanes. Both were claims-tier weighting/labeling discipline, not new stack elements.

— Cal A. Brate, Methodology Index v0.17 update Saturday 2026-06-06 ~10:30 EDT per Cal #256. σ -distance resolves the Finding-2 weight-map gap better than either v0.16 option; six-category framework retained as labeling layer over the σ metric; Refinement B adopted across lanes; seven-element stack stays closed.